Fractal Video: the equations

Every distance estimator in `vfx-apps/fractal-video/js/formulas.js`

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1 How to read this document

The program is a GPU ray marcher. It does *not* draw fractals by plotting points: for every pixel it marches a ray through space, and at each step asks a function DE(**p**) "how far is the nearest surface from here?". Every fractal in the program is therefore *one formula*: a **distance estimator** (DE) written in GLSL. This document translates all 34 of them into mathematics. For each one you get the iteration, the distance estimate, what the parameters do (with the app's default values), and a picture.

About the pictures. The renders are made by a small stand-alone ray marcher that loads the *same GLSL distance functions* from `formulas.js` with each formula's default parameters, default iteration count and default camera. The lighting and colouring are a simple stand-in (sun + soft shadow + orbit-trap colour), *not* the app's full look pipeline (palettes, AO, glow, tone-mapping), so colours will differ from what you see in the program. The shapes are exact.

1.1 The march

For a ray $\mathbf{r}(t) = \mathbf{o} + t\,\mathbf{d}$ the program steps $t \leftarrow t + \sigma\,\mathrm{DE}(\mathbf{r}(t))$ until $\mathrm{DE} < \varepsilon(t)$ (hit) or t is too large (miss). σ is the per-formula *step factor* (below 1 for estimators that overshoot).

1.2 Global space operations (applied before every formula)

Before the formula sees a point $\mathbf{p} = (x, y, z)$ the program may transform it (the *Space* tab). With twist rate τ, repeat period ρ and bound radius B:

$$a = \tau\, y, \qquad \mathbf{q} = \begin{pmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{pmatrix} \begin{pmatrix} x \\ z \end{pmatrix} \quad \text{(twist about the } y \text{ axis)} \tag{1}$$

$$\mathbf{q} \leftarrow \mathbf{q} - \rho\,\lfloor \mathbf{q}/\rho + \tfrac{1}{2} \rfloor \quad \text{(domain repetition, if } \rho > 0\text{)} \tag{2}$$

$$d_{\text{march}} = \sigma \cdot \max\big(\mathrm{DE}(\mathbf{q}),\ \|\mathbf{p}\| - B\big) \quad \text{(bounding sphere, if } B > 0\text{)} \tag{3}$$

1.3 Common building blocks

- **Rotations.** $\mathsf{R}_A, \mathsf{R}_B$ are rotation matrices from Euler angles (α, β, γ) in degrees, applied x then y then z: $\mathsf{R} = \mathsf{R}_z(\gamma)\mathsf{R}_y(\beta)\mathsf{R}_x(\alpha)$. Formulas with one rotation use R_A; those with two (R_A pre-, R_B post-rotation) say so.
- **Scale–offset map.** $S_{s,\mathbf{o}}(\mathbf{z}) = s\,\mathbf{z} - \mathbf{o}\,(s-1)$ (scale about the point $\mathbf{o}$).
- **Sort fold.** $\mathrm{sort}_{\downarrow}(\mathbf{z})$ permutes the components so that $z_x \geq z_y \geq z_z$ (three conditional swaps).
- **Box distance.** $\mathrm{sdBox}(\mathbf{q}, \mathbf{b}) = \|\max(|\mathbf{q}| - \mathbf{b},\, 0)\| + \min\big(\max(q_x, q_y, q_z),\, 0\big)$.
- **Running derivative.** Folds and scalings are conformal, so a scalar dr tracks the accumulated stretch and the final distance is divided by it.
- **Orbit trap** (for colouring only, never affects shape): $\mathbf{t} = \big(\min|z_x|, \min|z_y|, \min|z_z|, \min\|\mathbf{z}\|^2\big)$ over the orbit.

Sphere and box folds used by the Mandelbox family:

$$\mathrm{boxFold}_f(\mathbf{z}) = 2\ \mathrm{clamp}(\mathbf{z}, -f, f) - \mathbf{z}, \tag{4}$$

$$\mathrm{sphereFold}_{m,R}(\mathbf{z}, \mathrm{dr}): \quad k = \begin{cases} R^2/m^2 & \|\mathbf{z}\| < m \\ R^2/\|\mathbf{z}\|^2 & m \leq \|\mathbf{z}\| < R \\ 1 & \text{otherwise} \end{cases}, \quad \mathbf{z} \leftarrow k\mathbf{z}, \quad \mathrm{dr} \leftarrow k\,\mathrm{dr}. \tag{5}$$

And the *bulb power map* (spherical-coordinate nth power) used by the Mandelbulb family:

$$\mathcal{B}_n(\mathbf{z}) = r^n\big(\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta\big), \quad r = \|\mathbf{z}\|,\ \theta = n\arccos\frac{z_z}{r},\ \phi = n\ \mathrm{atan2}(z_y, z_x). \tag{6}$$

2 Escape-time fractals

2.1 Mandelbulb

Iterate with $\mathbf{z}_0 = \mathbf{p}$ and $\mathbf{c} = (1-j)\,\mathbf{p} + j\,\mathbf{c}_J$ (j = *Julia mix*, $\mathbf{c}_J$ = Julia constant):

$$r = \|\mathbf{z}_k\|, \quad \theta = n\arccos\frac{z_{k,z}}{r} + \theta_0, \quad \phi = n\ \mathrm{atan2}(z_{k,y}, z_{k,x}) + \phi_0, \tag{7}$$

$$\mathbf{z}_{k+1} = r^n\big(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta\big) + \mathbf{c}, \tag{8}$$

$$\mathrm{dr}_{k+1} = n\,r^{\,n-1}\,\mathrm{dr}_k + (1-j), \qquad \mathrm{dr}_0 = 1. \tag{9}$$

Stop when $\|\mathbf{z}\| > 2$ or after the iteration limit. The distance estimate (Hubbard–Douady potential)

$$\mathrm{DE}(\mathbf{p}) = \tfrac{1}{2}\,\frac{r\,\ln r}{\mathrm{dr}}. \tag{10}$$

n = *Power* (8 is the classic bulb), θ_0, ϕ_0 = phase offsets.

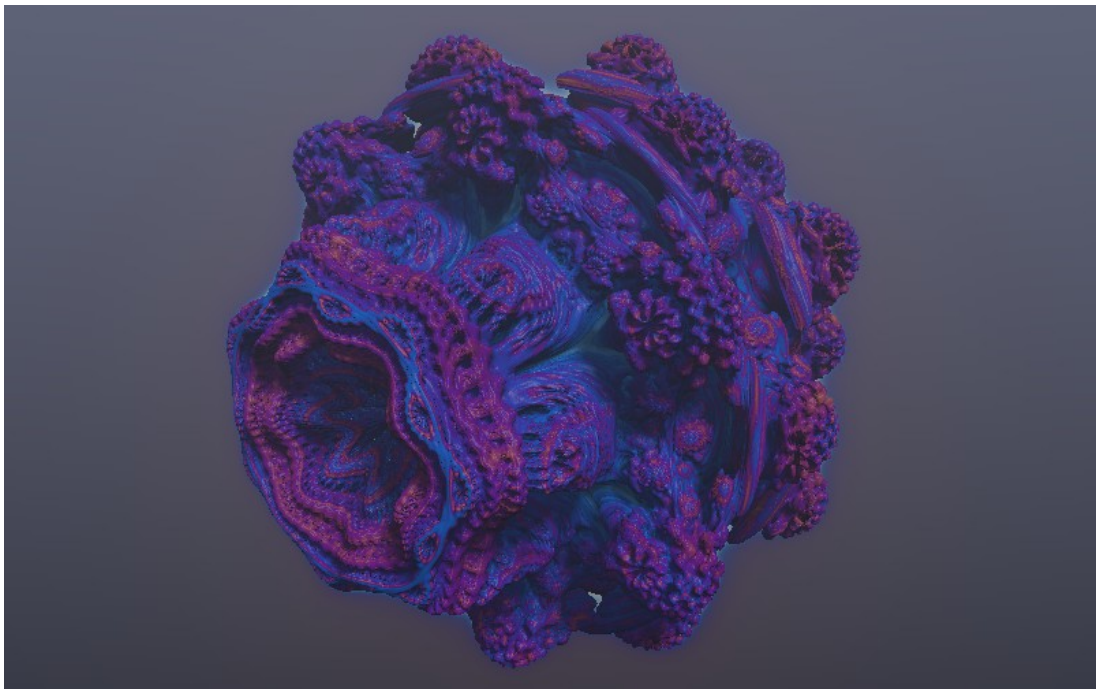


Figure 1: Mandelbulb, default parameters (8 iterations).

Parameters (app defaults). Iterations 8 (range 2–16); step factor 0.9; bounding size 1.2.

Parameter	Key	Default	Range
Power	`power`	8	2 . . . 16
Theta phase	`theta`	0°	−180 . . . 180
Phi phase	`phi`	0°	−180 . . . 180
Julia mix	`julia`	0	0 . . . 1
Julia c x	`cx`	0.35	−1.5 . . . 1.5
Julia c y	`cy`	0.3	−1.5 . . . 1.5
Julia c z	`cz`	−0.4	−1.5 . . . 1.5

2.2 Quaternion Julia

A 3D slice of the 4D Julia set of $q \mapsto q^2 + \mathbf{c}$ for quaternions. Write $\mathbf{z} = (z_w; \mathbf{v})$ with real part z_w and vector part $\mathbf{v} \in \mathbb{R}^3$. Start at $\mathbf{z}_0 = (p_x, p_y, p_z, w_0)$ where w_0 is the *4D slice* parameter:

$$\mathbf{z}_{k+1} = \left(z_x^2 - \|(z_y, z_z, z_w)\|^2,\ 2z_x\,(z_y, z_z, z_w)\right) + \mathbf{c}, \quad \mathbf{c} = (c_x, c_y, c_z, c_w), \tag{11}$$

$$\mathrm{md2}_{k+1} = 4\,\|\mathbf{z}_k\|^2\,\mathrm{md2}_k, \qquad \mathrm{md2}_0 = 1 \quad \text{(squared derivative norm)}, \tag{12}$$

$$\mathrm{DE} = \tfrac{1}{4}\sqrt{\frac{\|\mathbf{z}\|^2}{\mathrm{md2}}}\,\ln\|\mathbf{z}\|^2\,. \tag{13}$$

Escape when $\|\mathbf{z}\|^2 > 16$.

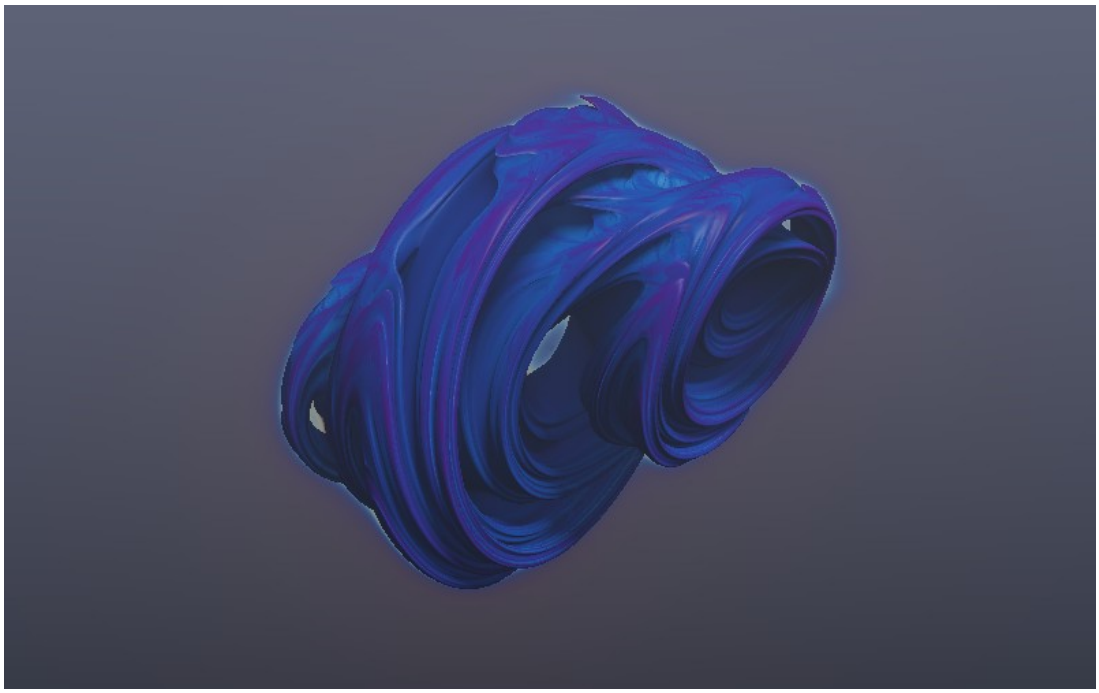


Figure 2: Quaternion Julia, default parameters (11 iterations).

Parameters (app defaults). Iterations 11 (range 2–20); step factor 0.9; bounding size 1.3.

Parameter	Key	Default	Range
c x	`cx`	-0.291	$-1.5 \dots 1.5$
c y	`cy`	-0.399	$-1.5 \dots 1.5$
c z	`cz`	0.339	$-1.5 \dots 1.5$
c w	`cw`	0.437	$-1.5 \dots 1.5$
4D slice	`slice`	0	$-1 \dots 1$

2.3 Folded bulb hybrid

A Mandelbulb whose orbit is mirror-folded and rotated before every e-th iteration (*Fold every*). With fold amount $\mu \in [0, 1]$:

$$\text{if } k \bmod e = 0: \quad \mathbf{z} \leftarrow \mathsf{R}_A\big((1-\mu)\mathbf{z} + \mu\,|\mathbf{z}|\,\big), \tag{14}$$

$$\mathrm{dr} \leftarrow n\,r^{n-1}\mathrm{dr} + 1, \qquad \mathbf{z} \leftarrow \mathcal{B}_n(\mathbf{z}) + \mathbf{p}, \tag{15}$$

with $r = \|\mathbf{z}\|$ evaluated after the fold. Escape at $r > 2$, and $\mathrm{DE} = \frac{1}{2}\, r \ln r / |\mathrm{dr}|$. $\mu = 0$ is the plain bulb; $\mu = 1$ is a full mirror fold.

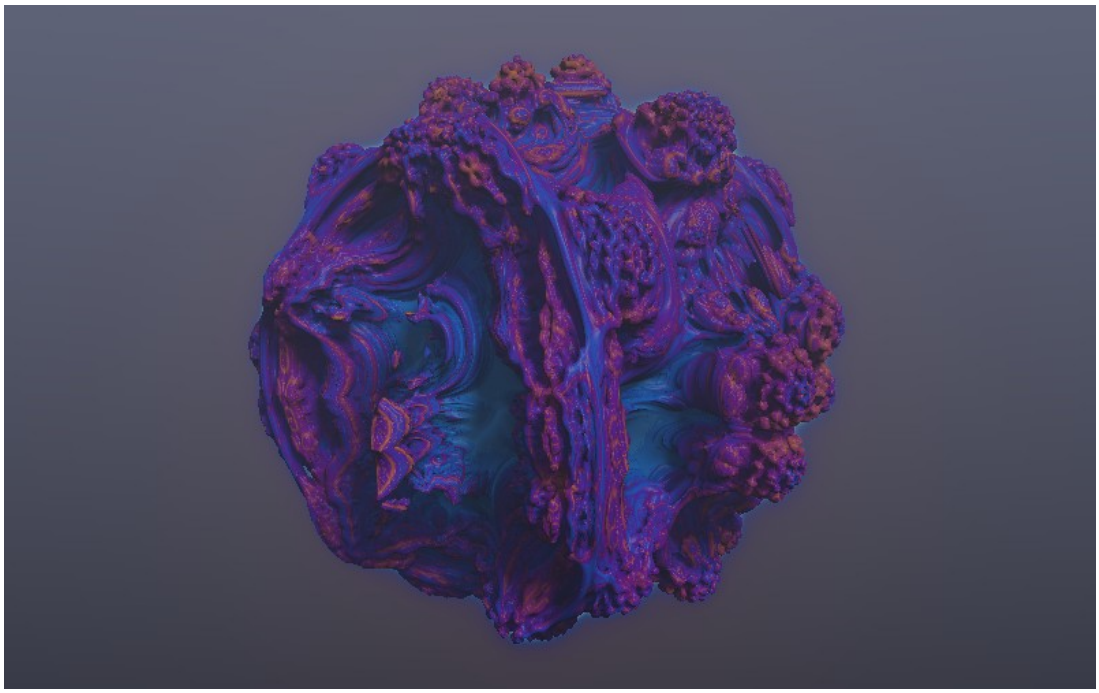


Figure 3: Folded bulb hybrid, default parameters (9 iterations).

Parameters (app defaults). Iterations 9 (range 2–20); step factor 0.85; bounding size 1.3.

Parameter	Key	Default	Range
Bulb power	`power`	8	2 . . . 12
Fold amount	`fold`	1	0 . . . 1
Fold every	`every`	1	1 . . . 4
Rotation X	`rotX`	0°	−180 . . . 180
Rotation Y	`rotY`	30°	−180 . . . 180
Rotation Z	`rotZ`	15°	−180 . . . 180

3 Mandelbox family (box fold + sphere fold)

All of these iterate $\mathbf{z} \leftarrow s\,\mathbf{z} + \mathbf{p}$ after folding, with $\mathrm{dr} \leftarrow |s|\ \mathrm{dr} + 1$, and return $\mathrm{DE} = \|\mathbf{z}\| \,/\, |\mathrm{dr}|$ (escape when $\|\mathbf{z}\|^2 > 10^4$). Negative s gives compact boxes, positive s open lattices.

3.1 Mandelbox

$$\mathbf{z} \leftarrow \mathrm{boxFold}_f(\mathbf{z}), \qquad (\mathbf{z}, \mathrm{dr}) \leftarrow \mathrm{sphereFold}_{m,R}(\mathbf{z}, \mathrm{dr}), \tag{16}$$

$$\mathbf{z} \leftarrow s\mathbf{z} + \mathbf{p}, \qquad \mathrm{dr} \leftarrow |s|\ \mathrm{dr} + 1, \tag{17}$$

with $m =$ *Min radius*, $R = \max(\text{Fixed radius}, m + 0.01)$, $f =$ *Fold limit*.

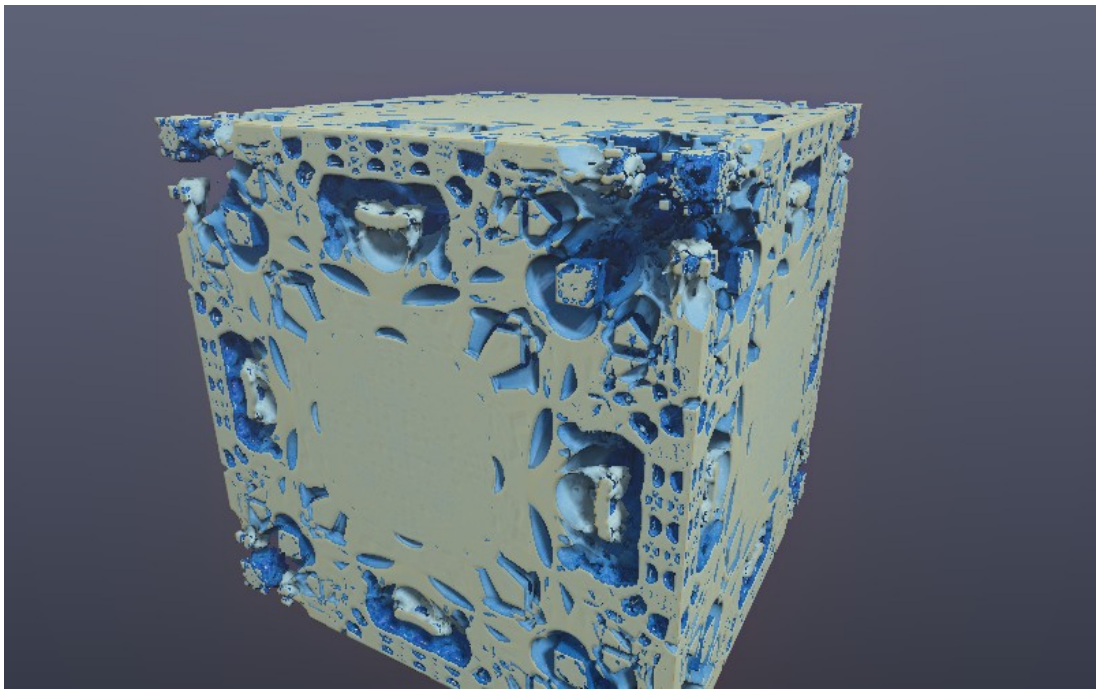


Figure 4: Mandelbox, default parameters (12 iterations).

Parameters (app defaults). Iterations 12 (range 2–24); step factor 1; bounding size 3.

Parameter	Key	Default	Range
Scale	`scale`	-1.8	$-3 \dots 3$
Min radius	`minR`	0.5	$0.05 \dots 1$
Fixed radius	`fixedR`	1	$0.3 \dots 2$
Fold limit	`fold`	1	$0.3 \dots 2$

3.2 Amazing Box

The Mandelbox with fixed $R = 1$ and a rotation inserted each iteration (organic, surface-like folds):

$$\mathbf{z} \leftarrow \mathsf{R}_A \, \mathrm{sphereFold}_{m,1}\big(\mathrm{boxFold}_f(\mathbf{z})\big), \qquad \mathbf{z} \leftarrow s\mathbf{z} + \mathbf{p}. \tag{18}$$

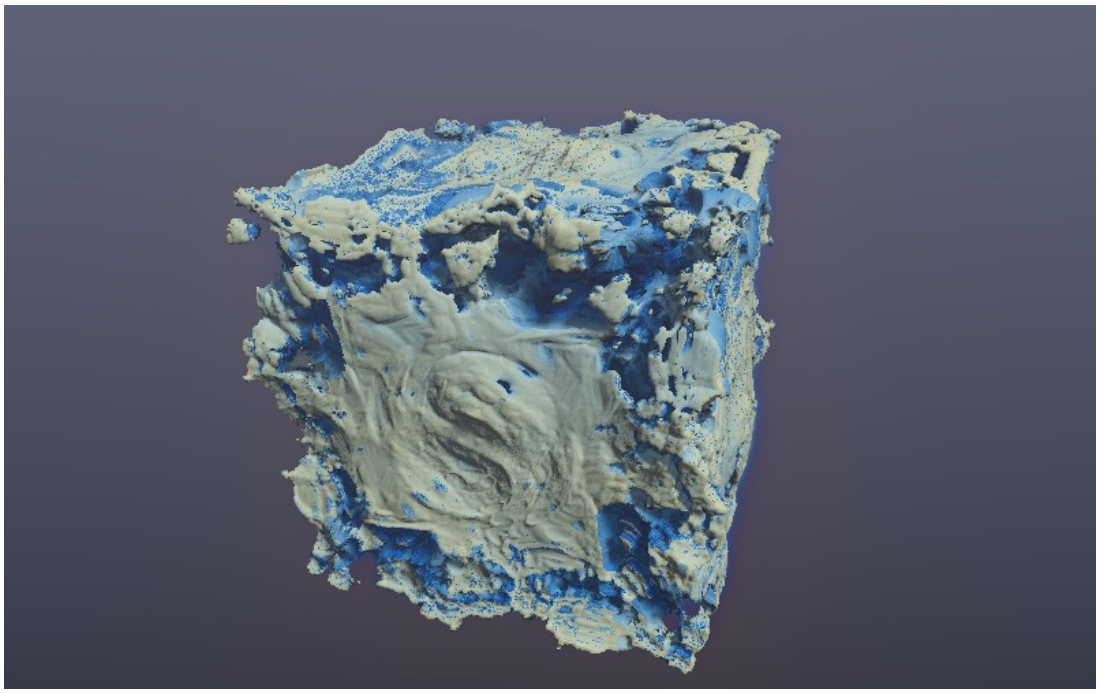


Figure 5: Amazing Box, default parameters (12 iterations).

Parameters (app defaults). Iterations 12 (range 2–24); step factor 0.8; bounding size 2.4.

Parameter	Key	Default	Range
Scale	`scale`	-1.7	$-3 \dots 3$
Min radius	`minR`	0.35	$0.05 \dots 1$
Fold limit	`fold`	1	$0.3 \dots 2$
Rotation X	`rotX`	12°	$-180 \dots 180$
Rotation Y	`rotY`	0°	$-180 \dots 180$
Rotation Z	`rotZ`	20°	$-180 \dots 180$

3.3 Vaulted box

Mandelbox with independent fold limits on the horizontal (f) and vertical (f_y) axes, and a sphere fold whose radius is blended towards a *cylinder* along z by the vault amount $v \in [0, 1]$:

$$\mathbf{z}_{xz} \leftarrow \mathrm{boxFold}_f(\mathbf{z}_{xz}), \quad z_y \leftarrow \mathrm{boxFold}_{f_y}(z_y), \tag{19}$$

$$r^2 = (1 - v)\ \|\mathbf{z}\|^2 + v\ \|\mathbf{z}_{xy}\|^2, \qquad k = \begin{cases} 1/m^2 & r < m \\ 1/r^2 & m \le r < 1 \\ 1 & \text{else} \end{cases}, \tag{20}$$

$$\mathbf{z} \leftarrow \mathsf{R}_A(k\mathbf{z}), \quad \mathrm{dr} \leftarrow k\,\mathrm{dr}, \qquad \mathbf{z} \leftarrow s\mathbf{z} + \mathbf{p}. \tag{21}$$

$v = 0$ is the Mandelbox; $v = 1$ gives barrel vaults running along z.

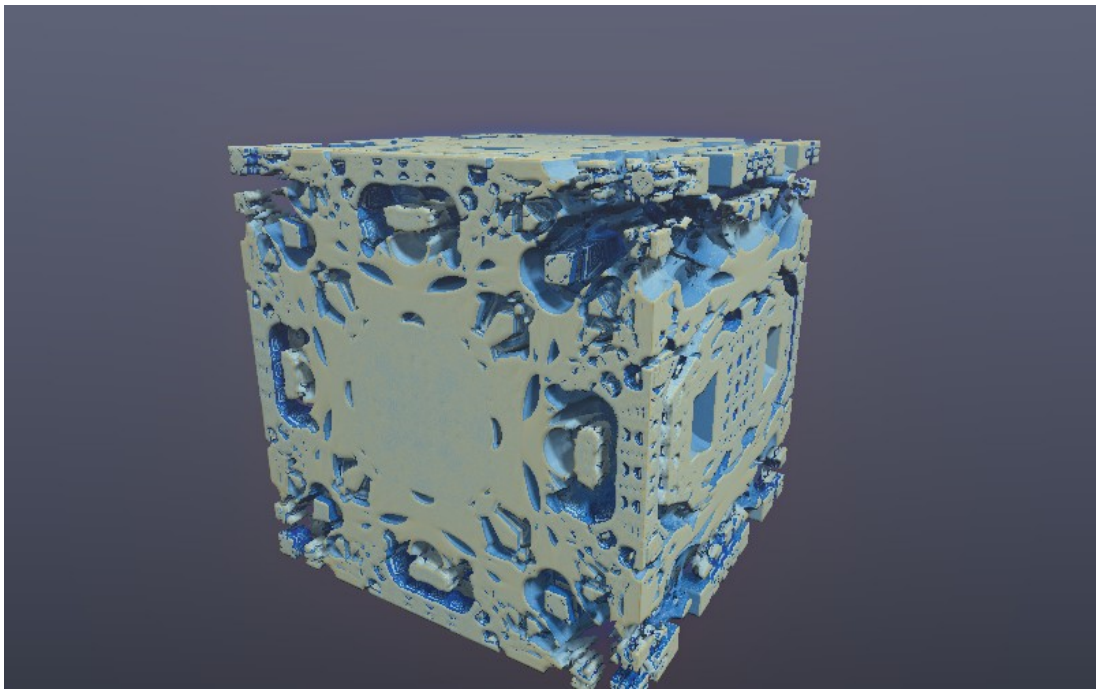


Figure 6: Vaulted box, default parameters (12 iterations).

Parameters (app defaults). Iterations 12 (range 2–24); step factor 0.8; bounding size 3.

Parameter	Key	Default	Range
Scale	`scale`	-1.8	$-3 \dots 3$
Min radius	`minR`	0.45	$0.05 \dots 1$
Fold limit	`fold`	1	$0.3 \dots 2$
Height fold	`foldY`	1	$0.3 \dots 2$
Vault	`vault`	0.8	$0 \dots 1$
Rotation X	`rotX`	0°	$-180 \dots 180$
Rotation Y	`rotY`	0°	$-180 \dots 180$
Rotation Z	`rotZ`	0°	$-180 \dots 180$

3.4 Amazing Surf

Like Amazing Box but the box fold acts on x, z only, leaving y unfolded, and an additive lift ℓ on y bends the sheets:

$$\mathbf{z}_{xz} \leftarrow \mathrm{boxFold}_f(\mathbf{z}_{xz}), \qquad (\mathbf{z}, \mathrm{dr}) \leftarrow \mathrm{sphereFold}_{m,1}(\mathbf{z}, \mathrm{dr}), \tag{22}$$

$$\mathbf{z} \leftarrow s\,\mathsf{R}_A\mathbf{z} + \mathbf{p} + (0, \ell, 0)^\top. \tag{23}$$

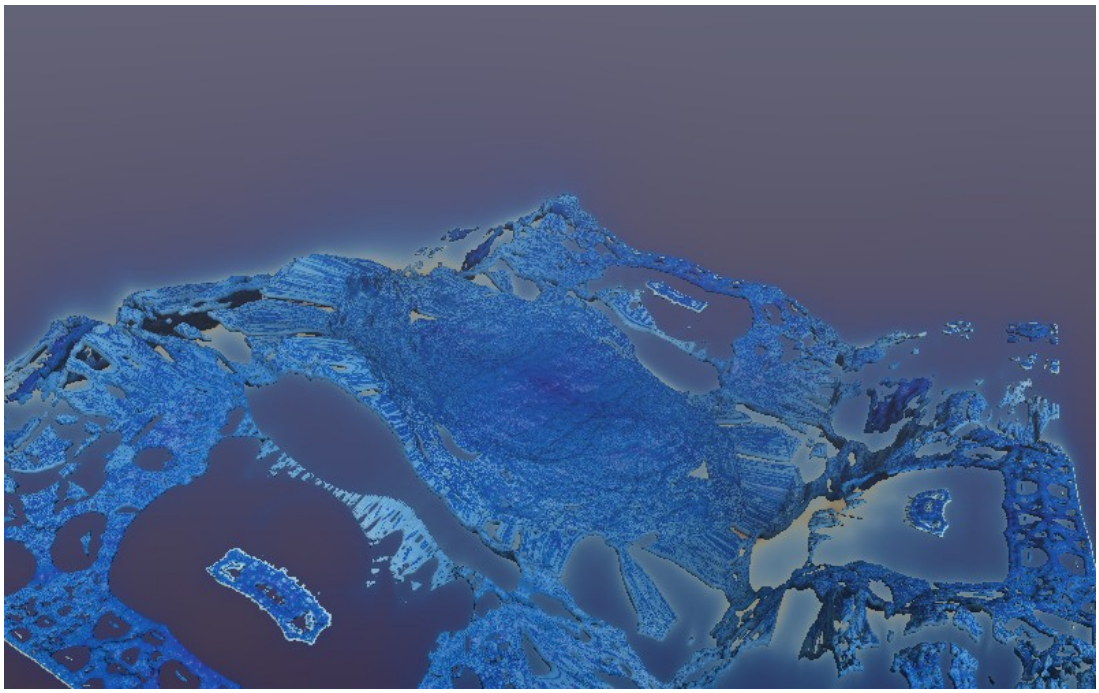


Figure 7: Amazing Surf, default parameters (10 iterations).

Parameters (app defaults). Iterations 10 (range 2–20); step factor 0.6; bounding size 1.6; *unbounded* (no bounding sphere assumed).

Parameter	Key	Default	Range
Scale	`scale`	-1.75	$-3 \dots 3$
Min radius	`minR`	0.45	$0.05 \dots 1$
Fold limit	`fold`	1	$0.3 \dots 2$
Sheet lift	`lift`	0	$-1 \dots 1$
Rotation X	`rotX`	6°	$-180 \dots 180$
Rotation Y	`rotY`	0°	$-180 \dots 180$
Rotation Z	`rotZ`	9°	$-180 \dots 180$

3.5 KaliBox

Kali's abs-fold set, carried as a 4-vector $\mathbf{Z} = (\mathbf{z}, w)$ with $w_0 = 1$ tracking the scale. With $m = \textit{Min radius}^2$, shift $\mathbf{t}$, Julia offset $\mathbf{c} = (c_x, c_y, c_z, 0)$:

$$\mathbf{z} \leftarrow |\mathbf{z}| + \mathbf{t}, \qquad r^2 = \|\mathbf{z}\|^2 , \tag{24}$$

$$\mathbf{Z} \leftarrow \mathbf{Z} \cdot \min\bigl(1, \max(m/r^2,\, m)\bigr) \quad \Bigl(= 1 \text{ if } r^2 \le m;\ m/r^2 \text{ if } m < r^2 < 1;\ m \text{ if } r^2 \ge 1 \Bigr), \tag{25}$$

$$\mathbf{Z} \leftarrow \frac{s}{m}\mathbf{Z} + \mathbf{c}, \qquad \mathbf{z} \leftarrow \mathsf{R}_A \mathbf{z}. \tag{26}$$

After N iterations:

$$\mathrm{DE} = \frac{\|\mathbf{z}\| - |s-1|}{|w|} - |s|^{\,1-N} . \tag{27}$$

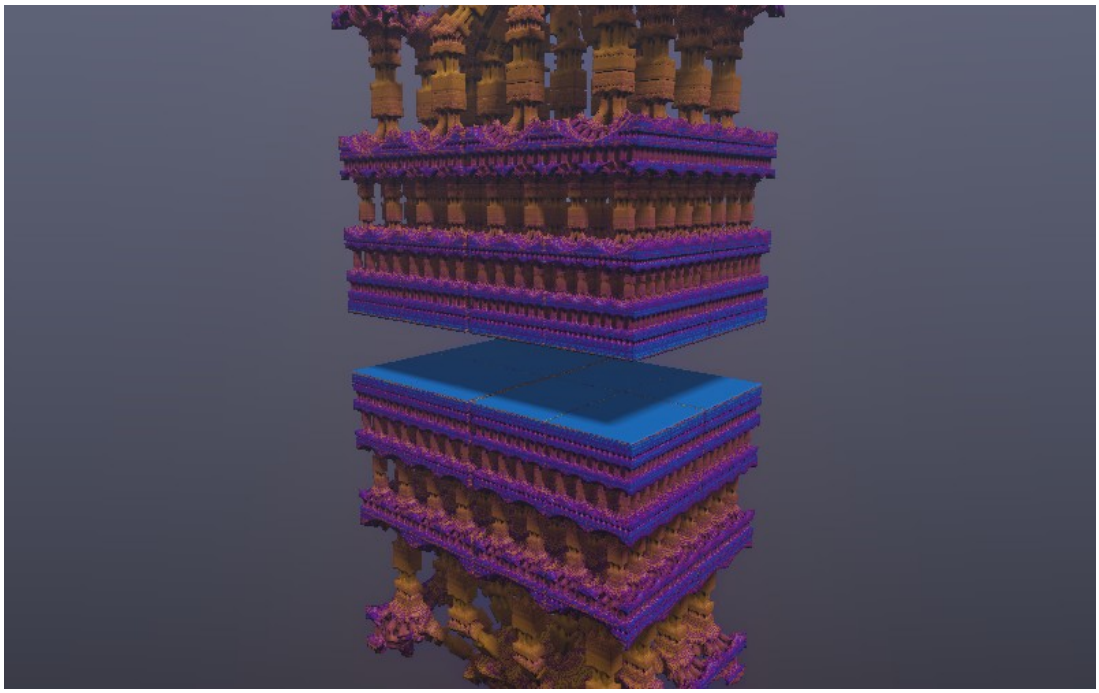


Figure 8: KaliBox, default parameters (13 iterations).

Parameters (app defaults). Iterations 13 (range 2–24); step factor 0.7; bounding size 2.2.

Parameter	Key	Default	Range
Scale	`scale`	2.043	1.2 … 3
Min radius^2	`minR2`	0.349	0.05 … 1
Shift X	`tx`	0.036	−2 … 2
Shift Y	`ty`	−1.861	−2.5 … 1
Shift Z	`tz`	0.036	−2 … 2
Julia X	`jx`	−0.669	−1.5 … 1.5
Julia Y	`jy`	−1.303	−1.5 … 1.5
Julia Z	`jz`	−0.458	−1.5 … 1.5
Rotation X	`rotX`	0°	−180 … 180
Rotation Y	`rotY`	0°	−180 … 180
Rotation Z	`rotZ`	0°	−180 … 180

3.6 Pseudo-Kleinian

Knighty's limit set of a Kleinian-like group: a box fold with half-extents $\mathbf{c}_s = (a, a, b)$ followed by a one-sided sphere inversion of size S and a translation $\mathbf{c}$:

$$\mathbf{z} \leftarrow 2\,\mathrm{clamp}(\mathbf{z}, -\mathbf{c}_s, \mathbf{c}_s) - \mathbf{z}, \qquad r^2 = \|\mathbf{z}\|^2, \tag{28}$$

$$k = \max\!\big(S/r^2,\, 1\big), \quad \mathbf{z} \leftarrow k\mathbf{z} + \mathbf{c}, \quad \mathrm{dr} \leftarrow k\,\mathrm{dr}. \tag{29}$$

With $\rho = \|\mathbf{z}_{xy}\|$ and strut radius t:

$$\mathrm{DE} = \frac{\max\!\Big(\rho - t,\; \dfrac{|\rho\, z_z|}{\|\mathbf{z}\|}\Big)}{\mathrm{dr}}. \tag{30}$$

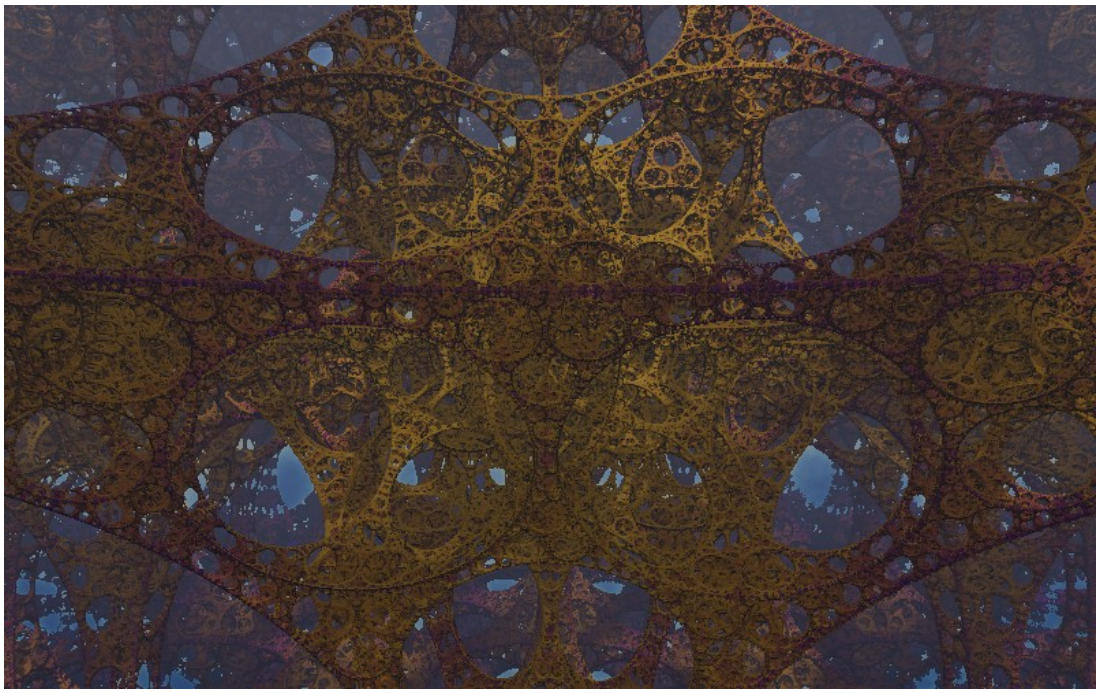


Figure 9: Pseudo-Kleinian, default parameters (9 iterations).

Parameters (app defaults). Iterations 9 (range 2–20); step factor 0.8; bounding size 1; *unbounded* (no bounding sphere assumed).

Parameter	Key	Default	Range
Box width	`boxXY`	0.908	0.4 . . . 1.4
Box height	`boxZ`	0.924	0.4 . . . 1.6
Inversion size	`size`	1	0.6 . . . 1.6
Offset X	`cx`	0	−0.5 . . . 0.5
Offset Y	`cy`	0	−0.5 . . . 0.5
Offset Z	`cz`	0	−0.5 . . . 0.5
Thickness	`thick`	0.928	0.5 . . . 1.2

3.7 Apollonian

Infinite sphere-inversion lattice. Per iteration, fold space into the cell $[-1,1]^3$ and invert:

$$\mathbf{z} \leftarrow -1 + 2\,\mathrm{fract}\big(\tfrac{1}{2}\mathbf{z} + \tfrac{1}{2}\big), \qquad r^2 = \|\mathbf{z}\|^2, \tag{31}$$

$$f = \frac{k}{\max(r^2, 10^{-6})}, \quad \mathbf{z} \leftarrow f\mathbf{z}, \quad \mathrm{sc} \leftarrow f\,\mathrm{sc}, \qquad \mathrm{DE} = \frac{|z_y|}{4\,\mathrm{sc}}. \tag{32}$$

k = *Inversion size*; the lattice is unbounded (use the bound radius to cut it out).

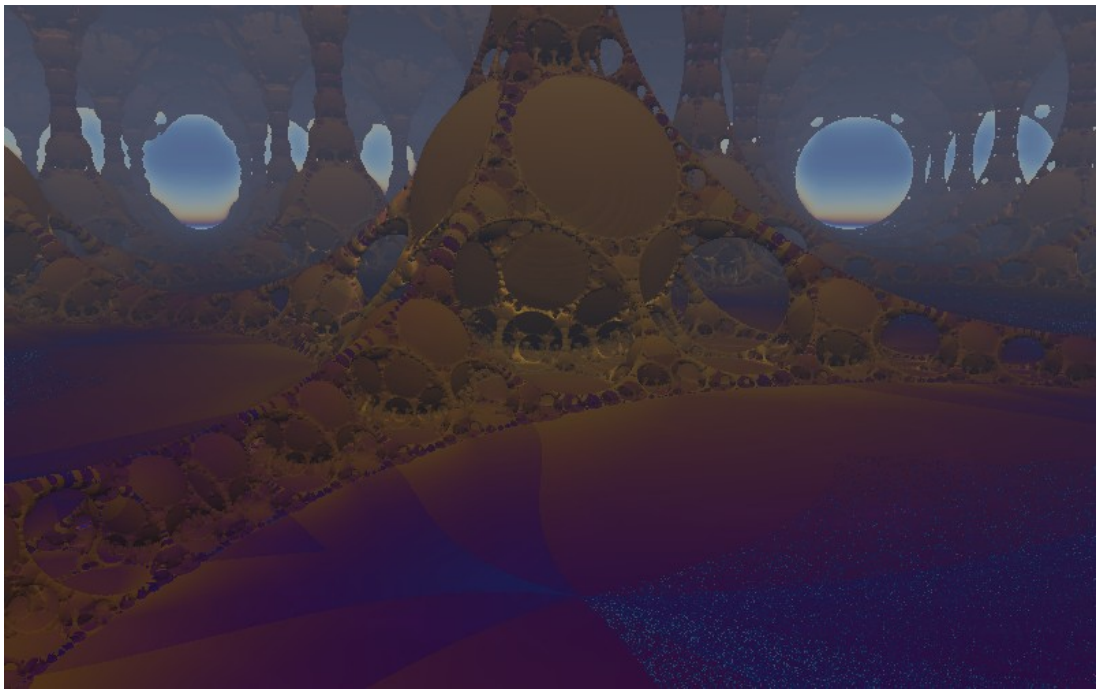


Figure 10: Apollonian, default parameters (8 iterations).

Parameters (app defaults). Iterations 8 (range 2–14); step factor 1; bounding size 1; *unbounded* (no bounding sphere assumed).

Parameter	Key	Default	Range
Inversion size	`k`	1.15	0.8 . . . 1.6

4 Kaleidoscopic IFS (fold, rotate, scale)

These are *iterated function systems*: instead of an escape test, each iteration applies a fold, a rotation and a scaling about an offset. They share $\mathrm{dr}_{k+1} = s\,\mathrm{dr}_k$, and the distance is that of a simple shape in the final folded frame, divided by dr.

4.1 Menger KIFS

$$\mathbf{z} \leftarrow \mathrm{sort}_{\downarrow}(|\mathbf{z}|), \qquad \mathbf{z} \leftarrow S_{s,\mathbf{o}}(\mathsf{R}_A \mathbf{z}), \tag{33}$$

$$\text{if } z_z < -\tfrac{1}{2} o_z (s-1): \quad z_z \leftarrow z_z + o_z(s-1), \tag{34}$$

$$\mathrm{DE} = \mathrm{sdBox}(\mathbf{z},\, (1,1,1)) \,/\, s^N. \tag{35}$$

With $s = 3$, $\mathbf{o} = (1,1,1)$, no rotation, this is the exact Menger sponge.

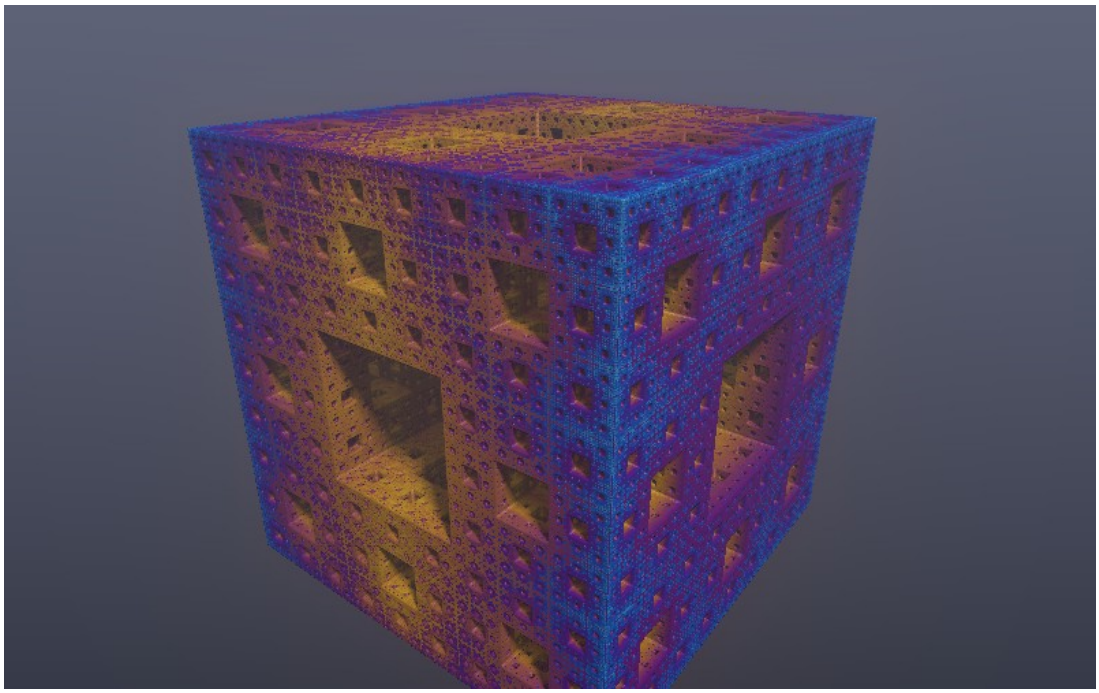


Figure 11: Menger KIFS, default parameters (6 iterations).

Parameters (app defaults). Iterations 6 (range 2–12); step factor 1; bounding size 1.4.

Parameter	Key	Default	Range
Scale	`scale`	3	2 . . . 4
Offset X	`offX`	1	0 . . . 2
Offset Y	`offY`	1	0 . . . 2
Offset Z	`offZ`	1	0 . . . 2
Rotation X	`rotX`	0°	−180 . . . 180
Rotation Y	`rotY`	0°	−180 . . . 180
Rotation Z	`rotZ`	0°	−180 . . . 180

4.2 Sierpinski

Tetrahedral folds (reflections in the planes $x+y=0$, $x+z=0$, $y+z=0$):

$$x+y<0:\ (x,y)\leftarrow(-y,-x);\quad x+z<0:\ (x,z)\leftarrow(-z,-x);\quad y+z<0:\ (y,z)\leftarrow(-z,-y), \tag{36}$$

$$\mathbf{z}\leftarrow S_{s,\mathbf{o}}(\mathsf{R}_A\mathbf{z}),\quad \mathbf{o}=(o,o,o),\qquad \mathrm{DE}=\big(\|\mathbf{z}\|-1\big)/s^N. \tag{37}$$

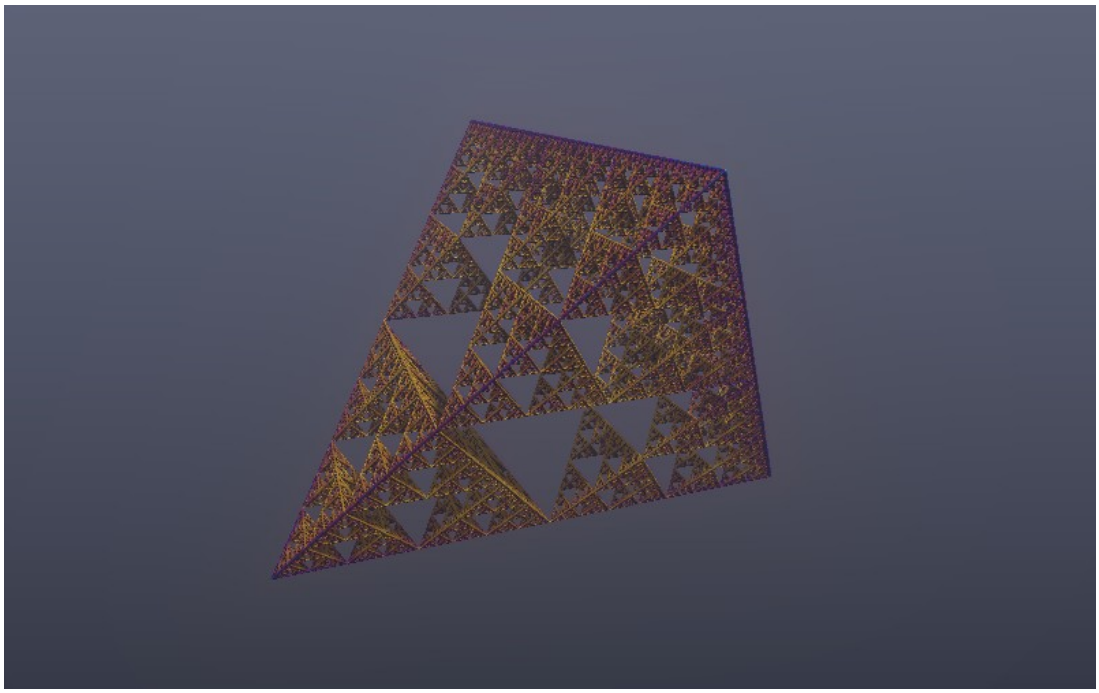


Figure 12: Sierpinski, default parameters (12 iterations).

Parameters (app defaults). Iterations 12 (range 2–20); step factor 1; bounding size 1.6.

Parameter	Key	Default	Range
Scale	`scale`	2	1.5 … 3
Offset	`offset`	1	0.3 … 1.6
Rotation X	`rotX`	0°	−180 … 180
Rotation Y	`rotY`	0°	−180 … 180
Rotation Z	`rotZ`	0°	−180 … 180

4.3 Octahedral KIFS

$$\mathbf{z} \leftarrow S_{s,\mathbf{o}}\big(\mathsf{R}_A\ \mathrm{sort}_{\downarrow}(|\mathbf{z}|)\big), \qquad \mathrm{DE} = \big(\,\|\mathbf{z}\| - 1\big)/s^N. \tag{38}$$

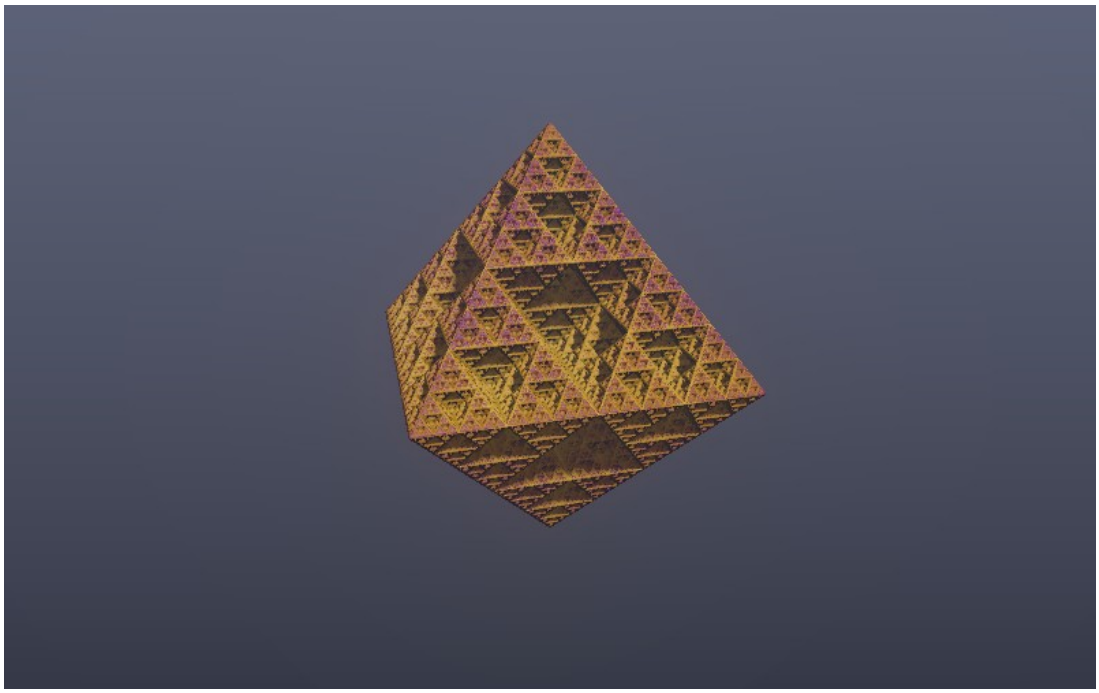


Figure 13: Octahedral KIFS, default parameters (12 iterations).

Parameters (app defaults). Iterations 12 (range 2–20); step factor 1; bounding size 1.8.

Parameter	Key	Default	Range
Scale	`scale`	2	1.5 … 3.5
Offset X	`offX`	1	0 … 2
Offset Y	`offY`	0	0 … 2
Offset Z	`offZ`	0	0 … 2
Rotation X	`rotX`	0°	−180 … 180
Rotation Y	`rotY`	0°	−180 … 180
Rotation Z	`rotZ`	0°	−180 … 180

4.4 Kaleidoscopic IFS

Two rotations around the same fold:

$$\mathbf{z} \leftarrow S_{s,\mathbf{o}}\Big(\mathsf{R}_B \ \mathrm{sort}_{\downarrow}\big(\,|\mathsf{R}_A\mathbf{z}|\,\big)\Big), \qquad \mathrm{DE} = \big(\,\|\mathbf{z}\| - 1\big)/s^N. \tag{39}$$

R_A = pre-rotation, R_B = post-rotation.

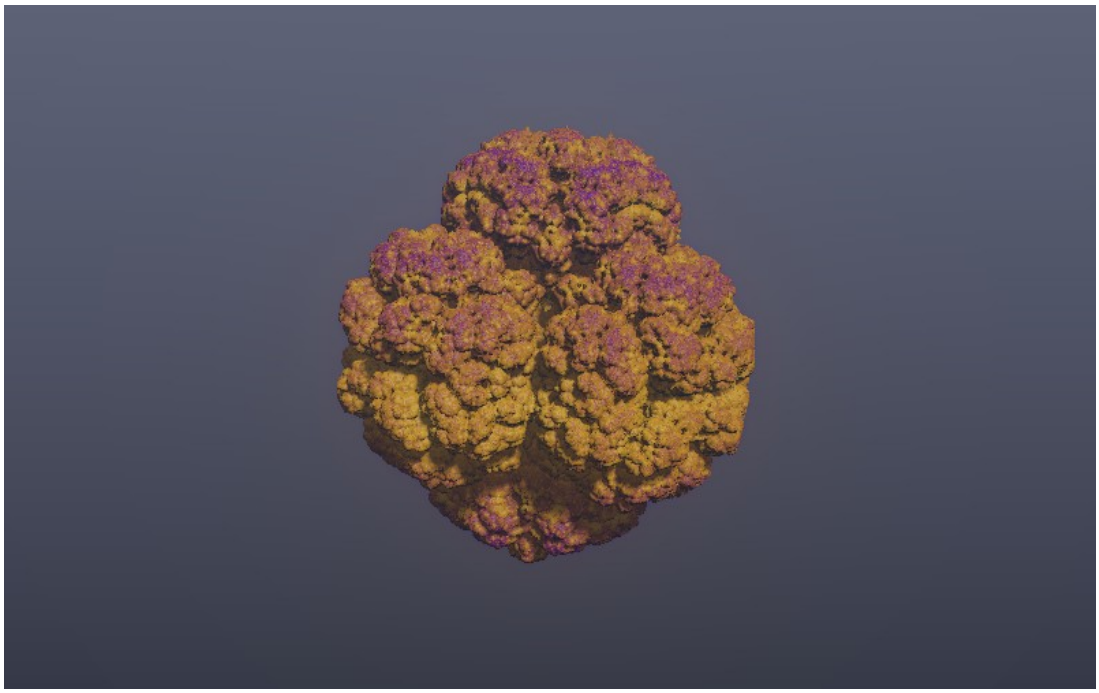


Figure 14: Kaleidoscopic IFS, default parameters (12 iterations).

Parameters (app defaults). Iterations 12 (range 2–20); step factor 0.9; bounding size 1.8.

Parameter	Key	Default	Range
Scale	`scale`	2.4	1.4 … 3.5
Offset X	`offX`	1	0 … 2
Offset Y	`offY`	0.5	0 … 2
Offset Z	`offZ`	0.25	0 … 2
Pre-rotation X	`preX`	10°	−180 … 180
Pre-rotation Y	`preY`	20°	−180 … 180
Pre-rotation Z	`preZ`	0°	−180 … 180
Post-rotation X	`postX`	0°	−180 … 180
Post-rotation Y	`postY`	0°	−180 … 180
Post-rotation Z	`postZ`	15°	−180 … 180

4.5 Icosahedral KIFS

Uses Knighty's polyhedral fold for icosahedral symmetry. Let $c = \cos(\pi/5) = 0.80901699\ldots$ and $\mathbf{n} = (-\frac{1}{2}, -c, \sqrt{\frac{3}{4} - c^2})$. The fold, applied five times, is

$$\mathcal{F}(\mathbf{z}): \quad \text{repeat 5 times: } \mathbf{z}_{xy} \leftarrow |\mathbf{z}_{xy}|, \quad \mathbf{z} \leftarrow \mathbf{z} - 2\min(0,\ \mathbf{z}\cdot\mathbf{n})\,\mathbf{n}. \tag{40}$$

Iteration:

$$\mathbf{z} \leftarrow S_{s,\mathbf{o}}\Big(\mathsf{R}_B\,\mathcal{F}(\mathsf{R}_A\mathbf{z})\Big), \qquad \mathrm{DE} = \big(\,\|\mathbf{z}\| - 1\big)/s^N. \tag{41}$$

The offset point $\mathbf{o}$ lies in the fundamental cell, spanned by a vertex $\mathbf{A} = (0, 0, 1)$, edge midpoint $\mathbf{B} \propto (\sigma, 0, \frac{1}{2})$ and face centre $\mathbf{C} \propto (0, \sigma, c)$ with $\sigma = \sqrt{\frac{3}{4} - c^2}$:

$$\mathbf{o} = \mathrm{reach} \cdot \frac{w_A\mathbf{A} + w_B\mathbf{B} + w_C\mathbf{C}}{\|w_A\mathbf{A} + w_B\mathbf{B} + w_C\mathbf{C}\|}. \tag{42}$$

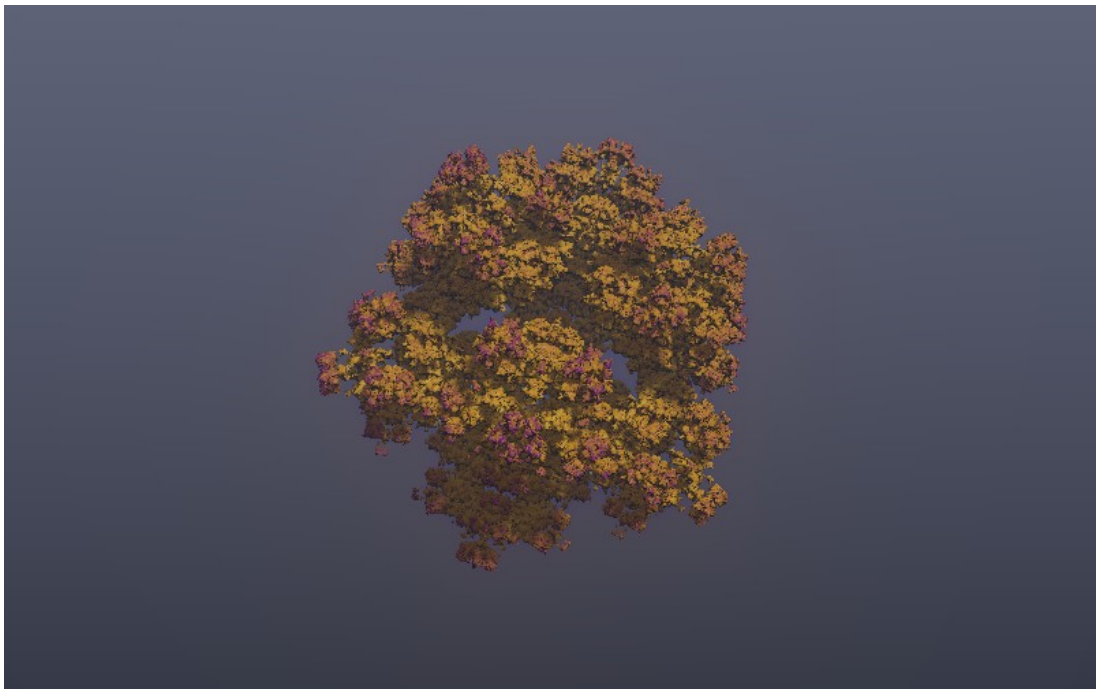


Figure 15: Icosahedral KIFS, default parameters (10 iterations).

Parameters (app defaults). Iterations 10 (range 2–20); step factor 0.9; bounding size 1.6.

Parameter	Key	Default	Range
Scale	`scale`	2.6	1.5 … 3.5
Vertex	`wA`	0	0 … 1
Edge	`wB`	1	0 … 1
Face	`wC`	0	0 … 1
Reach	`reach`	1	0.2 … 2
Pre-rotation X	`preX`	0°	−180 … 180
Pre-rotation Y	`preY`	8°	−180 … 180
Pre-rotation Z	`preZ`	0°	−180 … 180
Post-rotation X	`postX`	0°	−180 … 180
Post-rotation Y	`postY`	0°	−180 … 180
Post-rotation Z	`postZ`	12°	−180 … 180

4.6 Hyperbolic honeycomb

A $\{4,3,n\}$ honeycomb of cubes in the Poincaré ball model (here n = *Cubes per edge*, 5 for the compact $\{4,3,5\}$). Let $R^2 = 1/\cos(2\pi/n)$ be the squared radius of the cube's face spheres and $D = \sqrt{1+R^2}$ the distance of their centres, which are at $D\,\mathbf{e}_x,\ D\,\mathbf{e}_y,\ D\,\mathbf{e}_z$.

Travel/drift is a hyperbolic translation (Möbius map of the unit ball) that moves the origin to $\mathbf{a} = (\text{drift}, 0, \text{travel})$:

$$T_{\mathbf{a}}(\mathbf{x}) = \frac{(1 + 2\mathbf{a}\cdot\mathbf{x} + \|\mathbf{x}\|^2)\,\mathbf{a} + (1 - \|\mathbf{a}\|^2)\,\mathbf{x}}{1 + 2\mathbf{a}\cdot\mathbf{x} + \|\mathbf{a}\|^2\,\|\mathbf{x}\|^2}, \qquad \mathrm{sc}_0 = \frac{1 - \|\mathbf{a}\|^2}{1 + 2\mathbf{a}\cdot\mathbf{x} + \|\mathbf{a}\|^2\,\|\mathbf{x}\|^2}. \tag{43}$$

Then with $\mathbf{z}_0 = T_{\mathbf{a}}(\mathbf{p})$ and $\mathbf{c}_x = D\,\mathbf{e}_x$, each iteration folds into the fundamental region and inverts in the face sphere:

$$\mathbf{z} \leftarrow \mathrm{sort}_{\downarrow}(|\mathbf{z}|), \quad \mathbf{v} = \mathbf{z} - \mathbf{c}_x, \quad \text{stop if } \|\mathbf{v}\|^2 \geq R^2, \tag{44}$$

$$k = R^2/\|\mathbf{v}\|^2, \quad \mathbf{z} \leftarrow \mathbf{c}_x + k\mathbf{v}, \quad \mathrm{sc} \leftarrow k\,\mathrm{sc}. \tag{45}$$

The edges and vertices are drawn where the inversion spheres meet. With $f_i = \big|\,\|\mathbf{z} - D\mathbf{e}_i\| - R\big|$:

$$d = \min\Big(\,\|(f_x, f_y)\| - \text{edge},\ \|(f_x, f_y, f_z)\| - \text{node}\Big), \qquad \mathrm{DE} = \max\Big(\frac{d}{\mathrm{sc}},\ \|\mathbf{p}\| - \text{horizon}\Big). \tag{46}$$

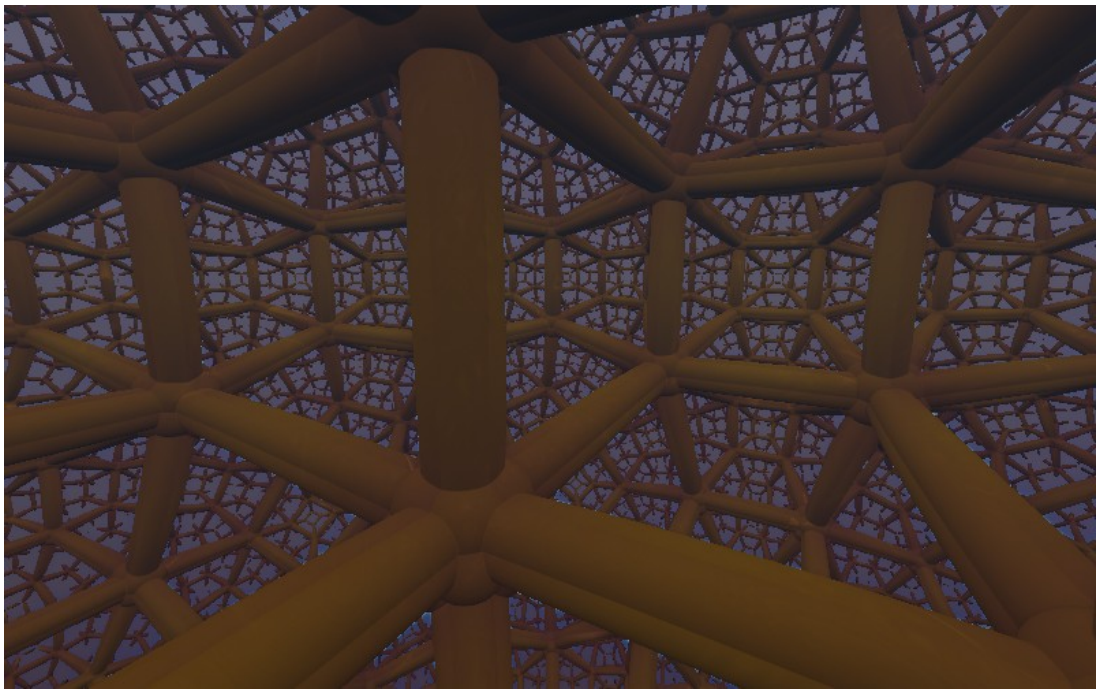


Figure 16: Hyperbolic honeycomb, default parameters (24 iterations).

Parameters (app defaults). Iterations 24 (range 4–32); step factor 0.8; bounding size 1.

Parameter	Key	Default	Range
Cubes per edge	`order`	5	4.2 … 8
Edge thickness	`edge`	0.045	0.005 … 0.2
Vertex size	`node`	0.08	0 … 0.3
Travel	`travel`	0	−0.95 … 0.95
Drift	`drift`	0	−0.95 … 0.95
Horizon	`shell`	0.985	0.8 … 1

5 Branching growth (tree, dendrite, nerve cell, root, mycelium, coral)

Six formulas share *one* estimator; they differ only in default parameters and in four constants. Each is a *tube that splits, turns and shrinks.*

One generation. Let $\mathrm{cap}_{h,r}(\mathbf{z}) = \|\mathbf{z} - (0, \mathrm{clamp}(z_y, 0, h), 0)\| - r$ be the distance to a vertical capsule of length h and radius r. State: the point $\mathbf{z}$ in the current branch's frame, world-up $\mathbf{u}$ carried in that frame, accumulated scale dr, current radius r_k (initially $r_0 = R$).

1. **Measure the branch:** $d \leftarrow \min\big(d,\ \mathrm{cap}_{L,r_k}(\mathbf{z})/\mathrm{dr}\big)$.
2. **Stand at its tip:** $z_y \leftarrow z_y - L$.
3. **Whorl.** With N branches per node, sector $\varsigma = 2\pi/N$ and azimuth $\alpha = \mathrm{atan2}(z_z, z_x)$, rotate about y into one wedge: $w = \big((\alpha + \varsigma/2) \bmod \varsigma\big) - \varsigma/2$, $\mathbf{z}_{xz} \leftarrow \|\mathbf{z}_{xz}\|\,(\cos w, \sin w)$, and rotate $\mathbf{u}$ by the same angle $w - \alpha$. A single capsule now stands for N branches.
4. **Lean** by the branch angle ϑ about z, then **twist** by ψ about y: $\mathbf{z} \leftarrow \mathsf{R}_y(\psi)\mathsf{R}_z(\vartheta)\mathbf{z}$ (same for $\mathbf{u}$).
5. **Droop** δ: with axis $\mathbf{a} = \mathbf{e}_y \times \mathbf{u} / \|\mathbf{e}_y \times \mathbf{u}\|$ (skipped if $\mathbf{u} \parallel \mathbf{e}_y$), $\mathbf{z} \leftarrow \mathsf{R}_{\mathbf{a}}(\delta)\mathbf{z}$, $\mathbf{u} \leftarrow \mathsf{R}_{\mathbf{a}}(\delta)\mathbf{u}$ (Rodrigues rotation). This bends every branch towards world-down.
6. **Child frame:** $\mathbf{z} \leftarrow \mathbf{z}/\varsigma_h$, $\mathrm{dr} \leftarrow \mathrm{dr}/\varsigma_h$, $r_{k+1} = \kappa\, r_k$, with $\varsigma_h =$ *Shrink* (clamped ≥ 0.2) and $\kappa =$ *Taper*.

Because every step is a rotation or uniform scale (conformal), dividing by dr gives a true distance *as long as each subtree stays inside its wedge*; for wide angles it can overestimate, which is why the default step factor is 0.35 and the iteration limit is held at 10. An N-way, 11-generation tree has N^{11} twigs but costs only 11 capsule evaluations.

Variants.

- **Root system:** the same with $y \to -y$ at the start (a tree grown downward) and strong droop ("gravity").
- **Nerve cell:** fixed $\kappa = 0.95$, no droop, plus a soma sphere $\|\mathbf{p}\| - \rho_s$ and one long unbranched *axon* $\mathrm{cap}_{A,\,0.9R}(p_x, -p_y, p_z)$ downward; d is minimised with both.
- **Tree, dendrite, mycelium, coral:** taper and droop are the user parameters; they differ in the defaults tabulated below.

	Shrink	Angle ϑ	Twist ψ	N	Seg. L	Radius R	Taper κ	Droop δ
Tree	0.68	32°	72°	4	1.00	0.110	0.86	8°
Dendrite	0.78	64°	0°	4	1.15	0.060	1.00	0°
Nerve cell	0.80	44°	96°	3	0.95	0.028	0.95 (fixed)	0
Root system	0.72	54°	38°	5	0.90	0.075	0.92	30°, y mirrored
Mycelium	0.87	36°	118°	6	1.25	0.016	1.02	−4°
Coral	0.74	52°	34°	6	0.62	0.200	0.97	−14°

Table 1: Branching defaults per formula (generated from the source). Nerve cell also has soma 0.26 and axon 2.6.

5.1 Tree



Figure 17: Tree, default parameters (7 iterations).

Parameters (app defaults). Iterations 7 (range 3–10); step factor 0.35; bounding size 1.5.

Parameter	Key	Default	Range
Shrink	`shrink`	0.68	0.45 … 0.88
Branch angle	`angle`	32°	0 … 85
Twist	`twist`	72°	−180 … 180
Branches	`folds`	4	2 … 8
Segment	`len`	1	0.4 … 2
Thickness	`rad`	0.11	0.02 … 0.4
Taper	`taper`	0.86	0.5 … 1.2
Droop	`droop`	8°	−60 … 60

5.2 Dendrite

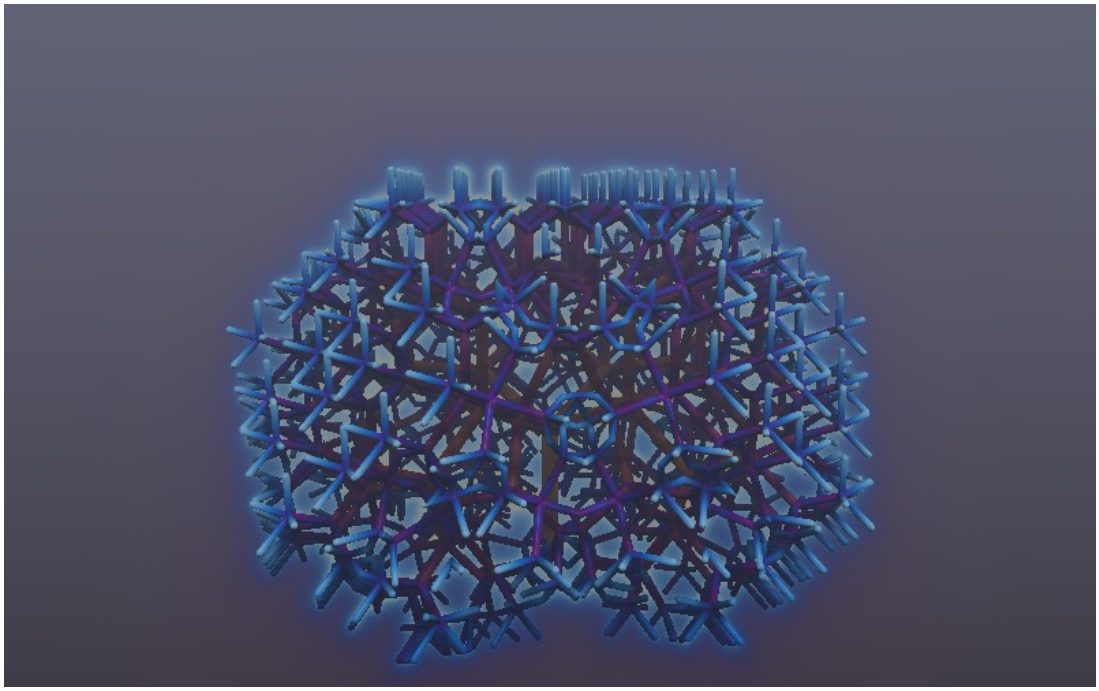


Figure 18: Dendrite, default parameters (7 iterations).

Parameters (app defaults). Iterations 7 (range 3–10); step factor 0.35; bounding size 1.5.

Parameter	Key	Default	Range
Shrink	`shrink`	0.78	0.45 … 0.92
Branch angle	`angle`	64°	0 … 85
Twist	`twist`	0°	−180 … 180
Branches	`folds`	4	2 … 8
Segment	`len`	1.15	0.4 … 2
Thickness	`rad`	0.06	0.02 … 0.4
Taper	`taper`	1	0.5 … 1.2
Droop	`droop`	0°	−60 … 60

5.3 Nerve cell

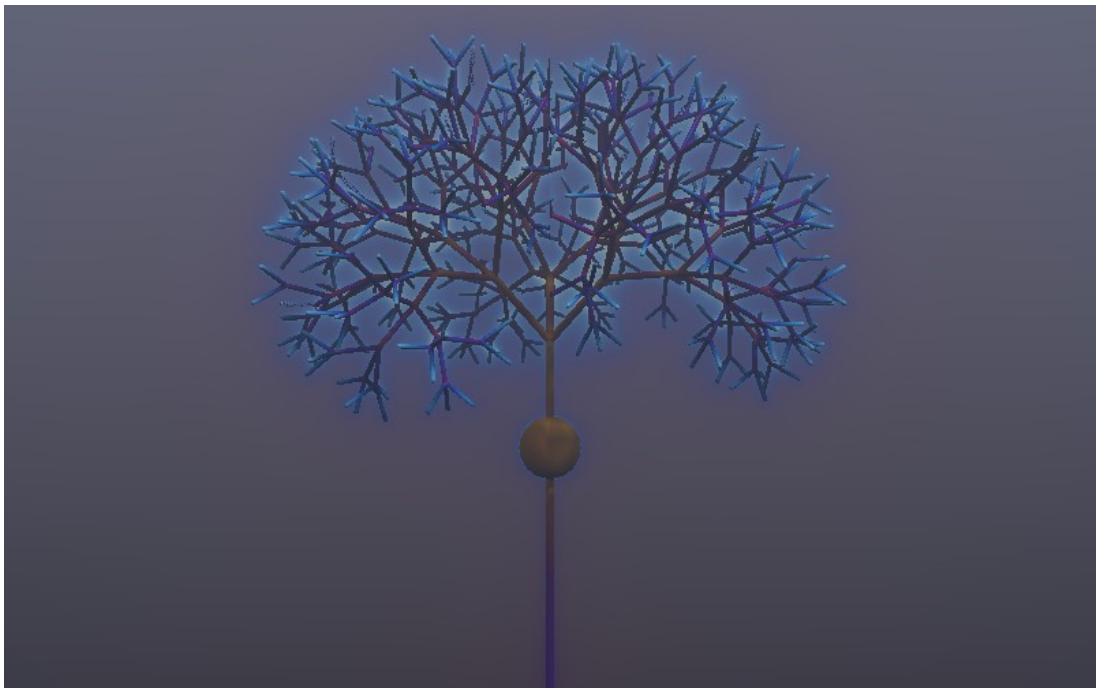


Figure 19: Nerve cell, default parameters (7 iterations).

Parameters (app defaults). Iterations 7 (range 4–10); step factor 0.35; bounding size 1.6.

Parameter	Key	Default	Range
Shrink	`shrink`	0.8	0.45 … 0.92
Branch angle	`angle`	44°	0 … 85
Twist	`twist`	96°	−180 … 180
Branches	`folds`	3	2 … 8
Segment	`len`	0.95	0.4 … 2
Thickness	`rad`	0.028	0.01 … 0.2
Cell body	`soma`	0.26	0 … 0.6
Axon	`axon`	2.6	0 … 6

5.4 Root system

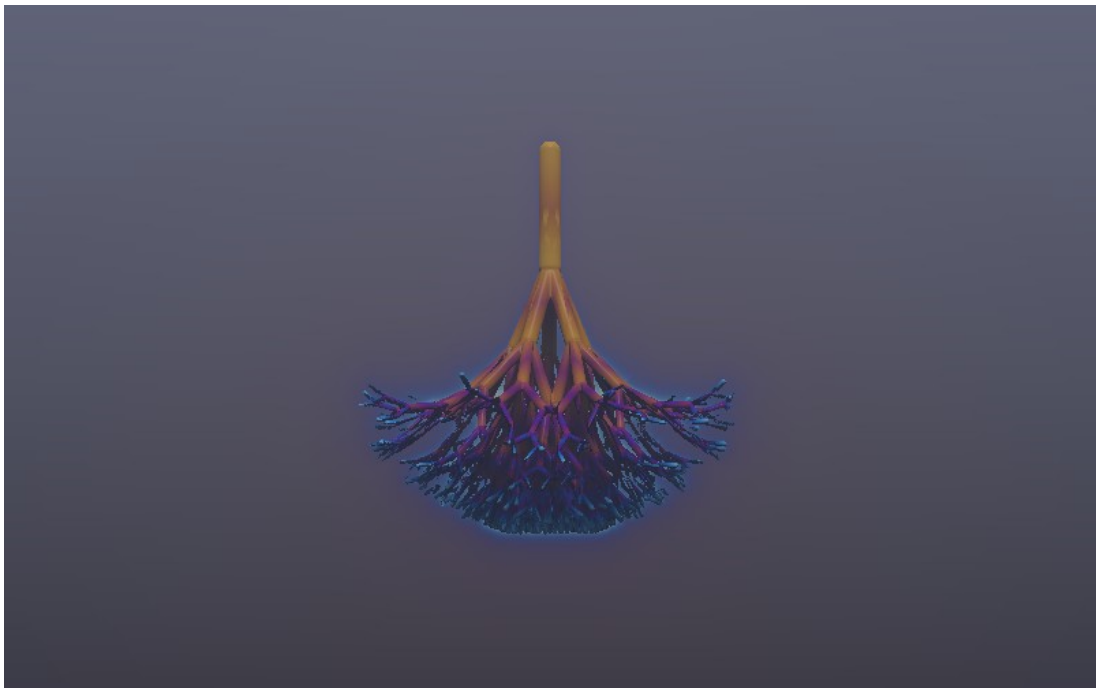


Figure 20: Root system, default parameters (7 iterations).

Parameters (app defaults). Iterations 7 (range 3–10); step factor 0.35; bounding size 1.5.

Parameter	Key	Default	Range
Shrink	`shrink`	0.72	0.45 … 0.9
Branch angle	`angle`	54°	0 … 85
Twist	`twist`	38°	−180 … 180
Branches	`folds`	5	2 … 8
Segment	`len`	0.9	0.4 … 2
Thickness	`rad`	0.075	0.02 … 0.4
Taper	`taper`	0.92	0.5 … 1.2
Gravity	`droop`	30°	−60 … 60

5.5 Mycelium

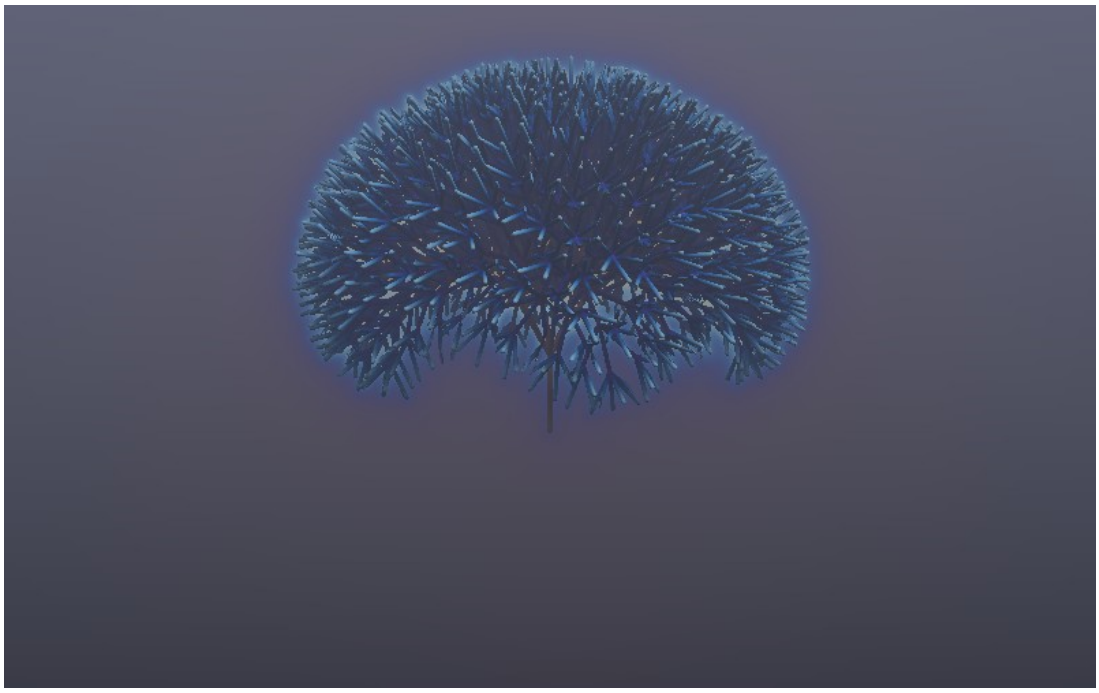


Figure 21: Mycelium, default parameters (7 iterations).

Parameters (app defaults). Iterations 7 (range 4–10); step factor 0.35; bounding size 1.8; *unbounded* (no bounding sphere assumed).

Parameter	Key	Default	Range
Shrink	`shrink`	0.87	0.55 … 0.95
Branch angle	`angle`	36°	0 … 85
Twist	`twist`	118°	−180 … 180
Branches	`folds`	6	2 … 8
Segment	`len`	1.25	0.4 … 2
Thickness	`rad`	0.016	0.005 … 0.1
Taper	`taper`	1.02	0.6 … 1.2
Drift	`droop`	−4°	−60 … 60

5.6 Coral

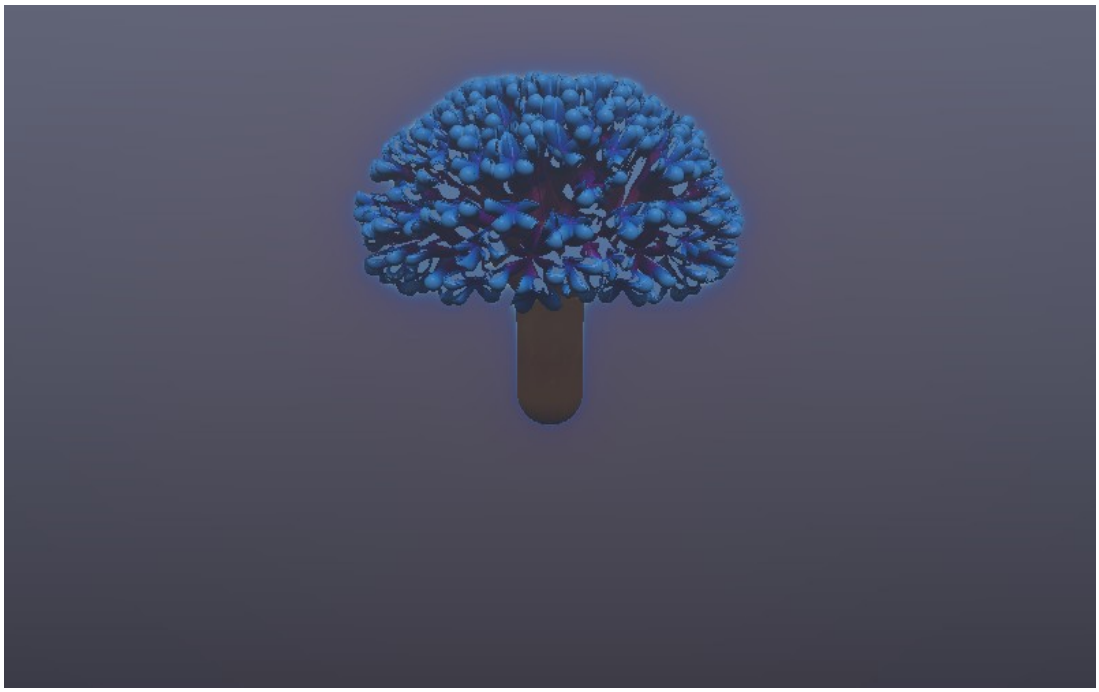


Figure 22: Coral, default parameters (6 iterations).

Parameters (app defaults). Iterations 6 (range 3–10); step factor 0.35; bounding size 1.5.

Parameter	Key	Default	Range
Shrink	`shrink`	0.74	0.45 … 0.92
Branch angle	`angle`	52°	0 … 85
Twist	`twist`	34°	−180 … 180
Branches	`folds`	6	2 … 8
Segment	`len`	0.62	0.4 … 2
Thickness	`rad`	0.2	0.02 … 0.4
Taper	`taper`	0.97	0.5 … 1.2
Droop	`droop`	−14°	−60 … 60

6 Architecture

6.1 Ziggurat

Four-way folded plan; each generation contributes a stepped pyramid of T tiers. With tier thickness $\vartheta =$ steepness$/T$ and half-width $w_i = 1 - 0.85\, i/T$:

$$\mathbf{z}_{xz} \leftarrow |\mathbf{z}_{xz}|\,, \quad \text{if } z_x < z_z : \text{ swap}, \tag{47}$$

$$p_d = \min_{i=0}^{T-1} \operatorname{sdBox}\Big(\mathbf{z} - (0,\ \vartheta(i+\tfrac{1}{2}),\ 0),\ \big(w_i,\ \tfrac{1}{2}\vartheta,\ w_i\big)\Big), \qquad d \leftarrow \min(d,\ p_d/\mathrm{dr}), \tag{48}$$

$$\mathbf{z} \leftarrow \mathsf{R}_y(\psi)\mathbf{z}, \quad \mathbf{z}_{xz} \leftarrow s\,\mathbf{z}_{xz} - \mathrm{spread}\,(s-1)(1,1), \quad z_y \leftarrow s\,z_y - \mathrm{rise}\,(s-1), \quad \mathrm{dr} \leftarrow s\,\mathrm{dr}. \tag{49}$$

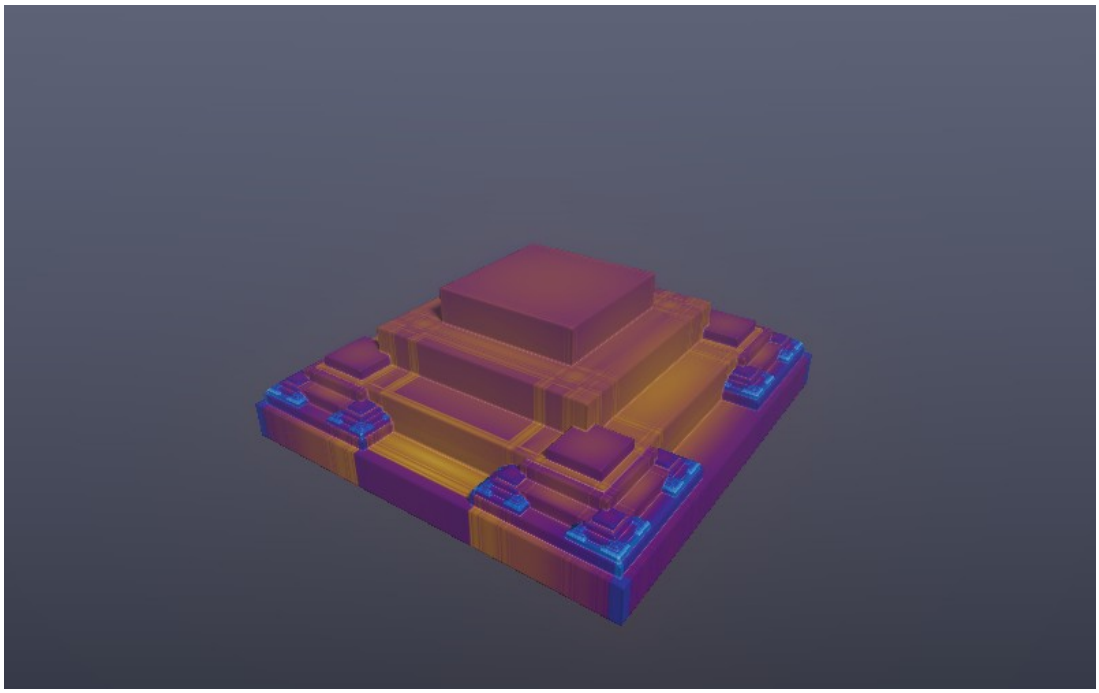


Figure 23: Ziggurat, default parameters (4 iterations).

Parameters (app defaults). Iterations 4 (range 1–8); step factor 0.8; bounding size 1.5.

Parameter	Key	Default	Range
Scale	`scale`	3	2 . . . 4
Spread	`spread`	0.95	0 . . . 2
Rise	`rise`	0.25	−1 . . . 1
Tiers	`tiers`	4	1 . . . 6
Steepness	`steep`	0.7	0.2 . . . 2
Twist	`rotY`	0°	−45 . . . 45

6.2 Skyline

Not a fractal: a procedural city. Space is cut into blocks of side c; block $\mathbf{i} = \lfloor p_{xz}/c \rfloor$ may hold a tower, decided by hashes $h_1, h_2, h_3 \in [0, 1)$ of $\mathbf{i} + 17 \cdot \text{seed}$, where $h(\mathbf{v}) = \text{fract}\big(P_x P_y\big)$ with $\mathbf{P} = \text{fract}(\mathbf{v} \odot (123.34, 456.21))$ followed by $\mathbf{P} \leftarrow \mathbf{P} + \mathbf{P} \cdot (\mathbf{P} + 45.32)$.

$$\text{tower exists} \iff h_2 < \text{density}, \qquad H = H_{\max}\big(1 - v + v\,1.6\,h_1^2\big), \tag{50}$$

$$w_x = \min\big(c\,\omega(0.85 + 0.3h_3),\, 0.45c\big), \quad w_z = \min\big(w'_x(0.75 + 0.5h_3),\, 0.45c\big), \tag{51}$$

(with ω the *tower width*, w'_x before clamping). The tower is a stack of T boxes (tier k of thickness $0.55H \cdot 0.6^k$, the last filling up to H), each narrowed by $(1 - \text{setback})^k$. The distance considers the block and its eight neighbours (towers never exceed their block): $d = \min\big(p_y,\ \min_{3\times3}\text{tower}\big)$, then clipped to the 3×3 patch edge by $\min(d,\ \min(1.5c - |q_x|\,,\ 1.5c - |q_z|) + 0.05c)$.

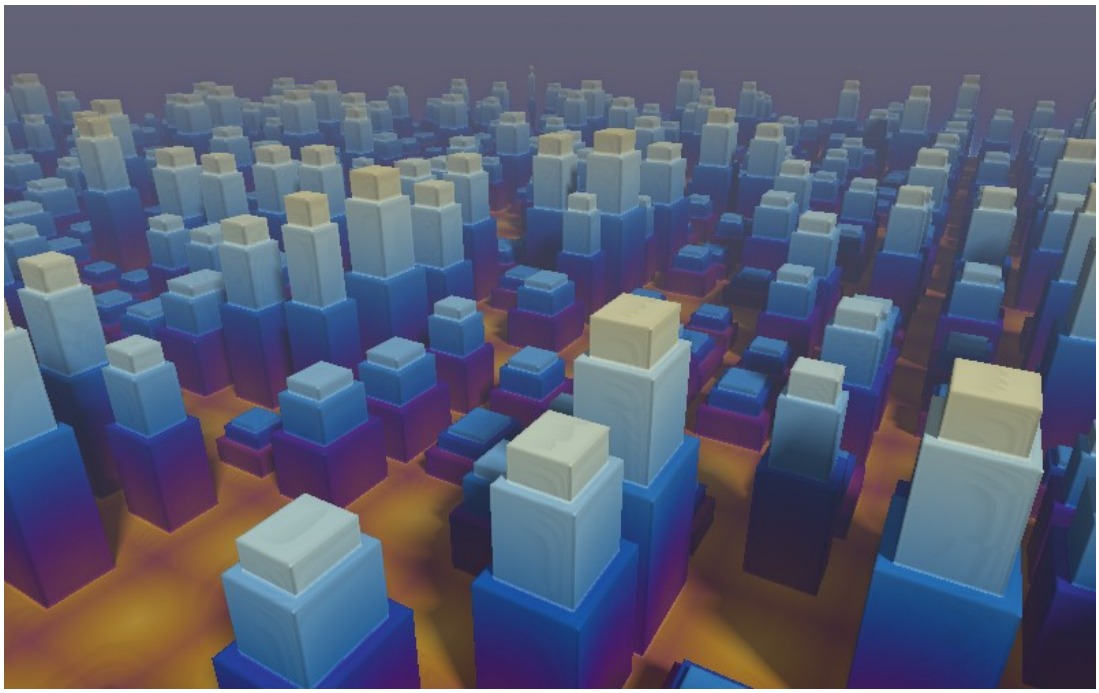


Figure 24: Skyline, default parameters (3 iterations).

Parameters (app defaults). Iterations 3 (range 1–5); step factor 0.8; bounding size 2; *unbounded* (no bounding sphere assumed).

Parameter	Key	Default	Range
Block size	`cell`	1	0.3 … 3
Height	`height`	1.8	0.1 … 6
Height variety	`vary`	0.75	0 … 1
Density	`density`	0.8	0 … 1
Tower width	`width`	0.32	0.15 … 0.42
Setback	`setback`	0.25	0 … 0.8
Seed	`seed`	7	0 … 100

7 Machines

7.1 Turbine

Procedural, not a fractal. S = iterations stages stacked along y at heights $y_k = (k + \frac{1}{2})g$ with radius $r_k = R\,\tau^k$ (g = gap, τ = taper). Blades per stage: N, sector $\varsigma = 2\pi/N$. For stage k:

$$a = \operatorname{atan2}(q_z, q_x) + \text{spin} \cdot (-1)^{k+1}, \qquad \mathbf{q} = \mathbf{p} - (0, y_k, 0) \quad \text{(alternate stages counter-rotate)}, \tag{52}$$

$$a' = a - \varsigma\lfloor a/\varsigma + \tfrac{1}{2}\rfloor, \qquad \mathbf{b} = \big(\rho\cos a',\ q_y,\ \rho\sin a'\big),\ \rho = \|\mathbf{q}_{xz}\|\,, \tag{53}$$

$$\mathbf{b}_{yz} \leftarrow \begin{pmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix} \mathbf{b}_{yz} \quad \text{(blade pitch } \varphi\text{)}, \tag{54}$$

$$d_{\text{blade}} = \text{sdBox}\Big(\mathbf{b} - \big(\tfrac{1}{2}(r_k + h), 0, 0\big),\ \big(\tfrac{1}{2}(r_k - h),\ 0.012 + 0.04 r_k,\ \chi\, r_k\big)\Big), \tag{55}$$

$$d_{\text{ring}} = \|(\rho - r_k - 0.03,\ q_y)\| - 0.025 r_k, \qquad d_{\text{stage}} = 0.9\min(d_{\text{blade}}, d_{\text{ring}}). \tag{56}$$

h = shaft radius, χ = blade width. The shaft is $\max\big(\|\mathbf{p}_{xz}\| - h,\ |p_y - \frac{1}{2}\ell| - \frac{1}{2}\ell - 0.15\big)$ with $\ell = gS$. The result is the minimum of shaft and all stages.

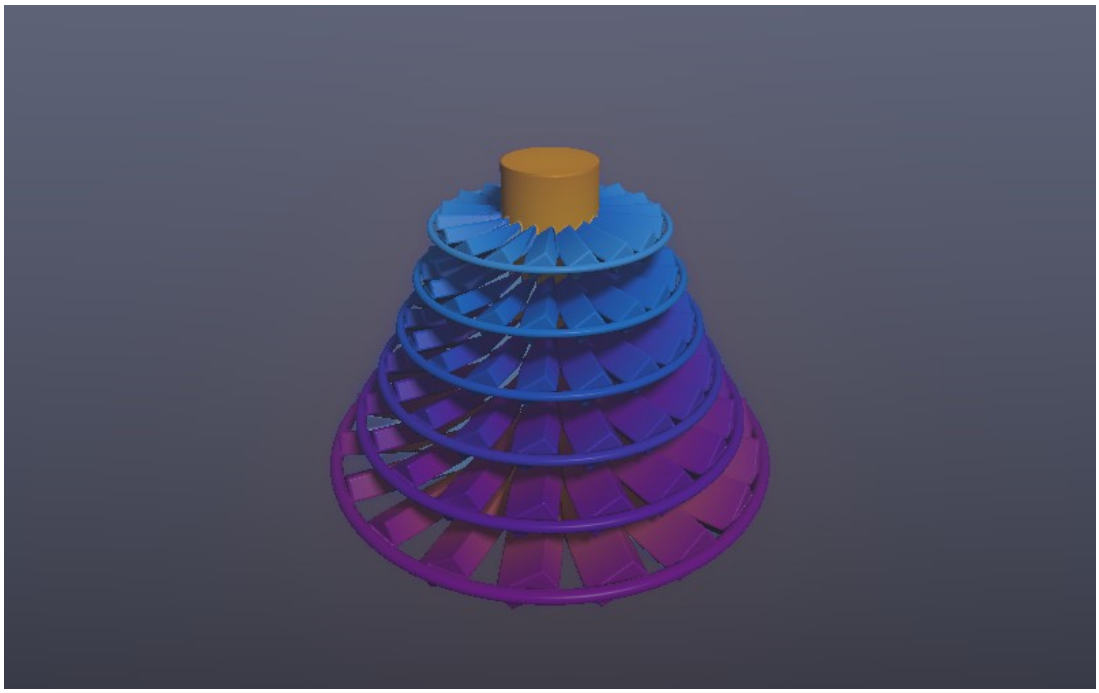


Figure 25: Turbine, default parameters (6 iterations).

Parameters (app defaults). Iterations 6 (range 1–12); step factor 0.7; bounding size 1.6.

Parameter	Key	Default	Range
Blades	`blades`	18	2 … 40
Radius	`radius`	1	0.3 … 2
Taper	`taper`	0.86	0.5 … 1
Stage gap	`gap`	0.22	0.05 … 0.6
Blade pitch	`pitch`	35°	0 … 80
Blade width	`chord`	0.11	0.02 … 0.4
Shaft	`hub`	0.2	0.02 … 0.6
Spin	`spin`	0°	−180 … 180

7.2 Greeble hull

Folded panels: every generation adds a plate, so plates carry plates. For each iteration (plate thickness t, seam depth ϵ):

$$\mathbf{z} \leftarrow \mathsf{R}_A \, \mathrm{fold}_3(|\mathbf{z}|) \quad (\text{swap } xy,\ yz,\ xy \text{ to sort}), \tag{57}$$

$$\mathbf{q} = \mathbf{z} - (0, 1 - t, 0), \qquad \mathbf{g} = \left|\mathrm{fract}(2\mathbf{q}_{xz}) - \tfrac{1}{2}\right|, \tag{58}$$

$$d_{\text{plate}} = \mathrm{sdBox}(\mathbf{q}, (1, t, 1)) + \epsilon \, \mathrm{smoothstep}\big(0.06, 0, \min(g_x, g_z)\big), \tag{59}$$

$$d \leftarrow \min(d,\ d_{\text{plate}}/\mathrm{dr}), \qquad \mathbf{z} \leftarrow S_{s,\mathbf{o}}(\mathbf{z}), \quad \mathrm{dr} \leftarrow s\,\mathrm{dr}. \tag{60}$$

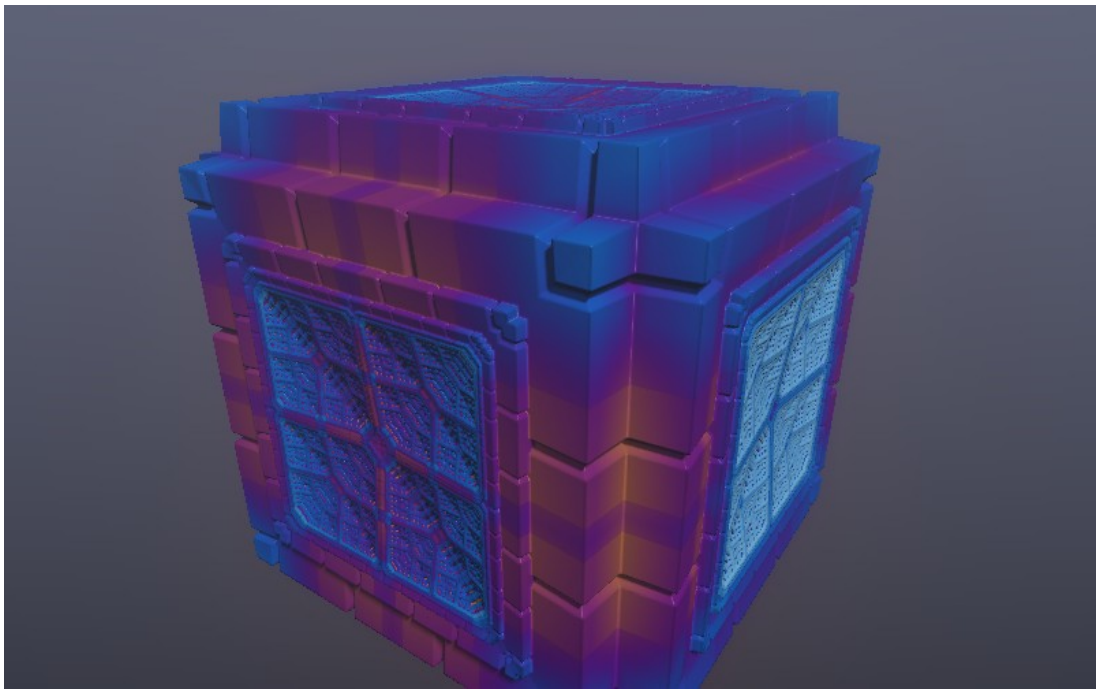


Figure 26: Greeble hull, default parameters (6 iterations).

Parameters (app defaults). Iterations 6 (range 1–12); step factor 0.8; bounding size 1.8.

Parameter	Key	Default	Range
Scale	`scale`	2.3	1.5 … 3.5
Offset X	`offX`	1	0 … 2
Offset Y	`offY`	0.7	0 … 2
Offset Z	`offZ`	0.45	0 … 2
Plate thickness	`plate`	0.22	0.05 … 0.8
Panel seams	`inset`	0.03	0 … 0.1
Rotation X	`rotX`	0°	−180 … 180
Rotation Y	`rotY`	0°	−180 … 180
Rotation Z	`rotZ`	6°	−180 … 180

7.3 Pipeworks

Each generation runs a pipe along x from the centre, with a flange at its far end, and the symmetric fold maps the other axes onto it. Pipe radius ρ_p, joint radius j, $L = 1$:

$$\mathbf{z} \leftarrow \mathsf{R}_A \,\mathrm{sort}_{\downarrow}(|\mathbf{z}|), \qquad d_{\text{pipe}} = \max\big(\,\|\mathbf{z}_{yz}\| - \rho_p,\; z_x - L\big), \tag{61}$$

$$\mathbf{f} = \big(\,\|\mathbf{z}_{yz}\| - j,\; |z_x - L| - 0.35\,j\big), \qquad d_{\text{fl}} = \min(\max(f_x, f_y), 0) + \|\max(\mathbf{f}, 0)\|\,, \tag{62}$$

$$d \leftarrow \min\big(d,\; \min(d_{\text{pipe}}, d_{\text{fl}})/\mathrm{dr}\big), \qquad \mathbf{z} \leftarrow S_{s,\mathbf{o}}(\mathbf{z}),\quad \mathrm{dr} \leftarrow s\,\mathrm{dr}. \tag{63}$$

With $s = 3$ and $o_x = 1.5$ each smaller cross sits exactly on a pipe end.

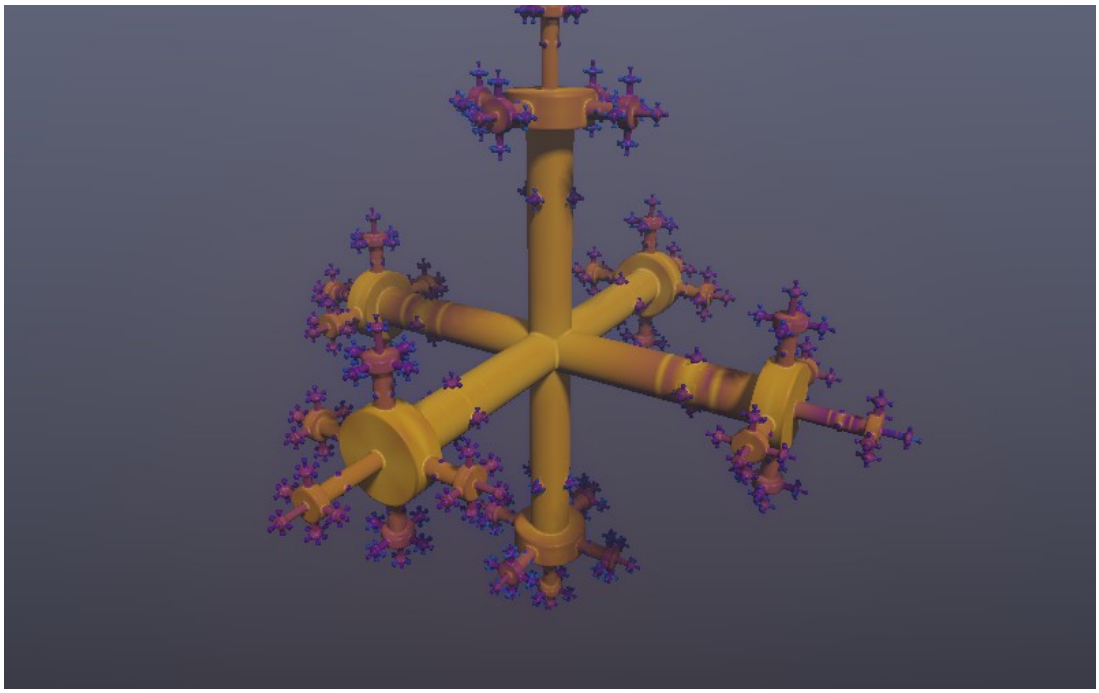


Figure 27: Pipeworks, default parameters (4 iterations).

Parameters (app defaults). Iterations 4 (range 1–10); step factor 0.8; bounding size 1.8.

Parameter	Key	Default	Range
Scale	`scale`	3	1.5 … 4
Offset X	`offX`	1.5	0 … 2
Offset Y	`offY`	0	0 … 2
Offset Z	`offZ`	0	0 … 2
Pipe radius	`pipe`	0.09	0.02 … 0.5
Joint size	`joint`	0.17	0 … 1
Rotation X	`rotX`	0°	−180 … 180
Rotation Y	`rotY`	0°	−180 … 180
Rotation Z	`rotZ`	0°	−180 … 180

7.4 Piston field

An endless periodic deck of telescoping pistons. Cell size c, block $\mathbf{i} = \lfloor p_{xz}/c \rfloor$, local coordinates $\mathbf{q}$ relative to the cell centre, rod radius $r_0 = \text{radius} \cdot c$. Phase and rod height:

$$\text{ph} = 2\pi\big(\phi + \omega \cdot 0.11\,(0.7\,i_x + 1.3\,i_z)\big), \qquad H = \max\big(h_0 + \Delta(\tfrac{1}{2} + \tfrac{1}{2}\sin\text{ph}),\, 0.02\big), \tag{64}$$

(ϕ = phase, ω = wave, h_0 = rest height, Δ = stroke). With the cylinder distance $\text{cyl}(\mathbf{q}; r, y_0, y_1) = \min(\max(e_x, e_y), 0) + \|\max(\mathbf{e}, 0)\|$, $\mathbf{e} = (\|\mathbf{q}_{xz}\| - r,\ \big|q_y - \tfrac{1}{2}(y_0 + y_1)\big| - \tfrac{1}{2}(y_1 - y_0))$, the pistons are S stacked stages

$$\text{cyl}\big(\mathbf{q};\ r_0(1 - 0.18k),\ kH/S - 0.01,\ (k+1)H/S\big),\ k = 0..S-1, \tag{65}$$

plus a cap $\text{cyl}(\mathbf{q};\, r_0(0.9 + 0.9\kappa),\, H,\, H + 0.12c(\tfrac{1}{2} + \kappa))$ and a deck with a collar round each rod; the result is clipped to the cell boundary.

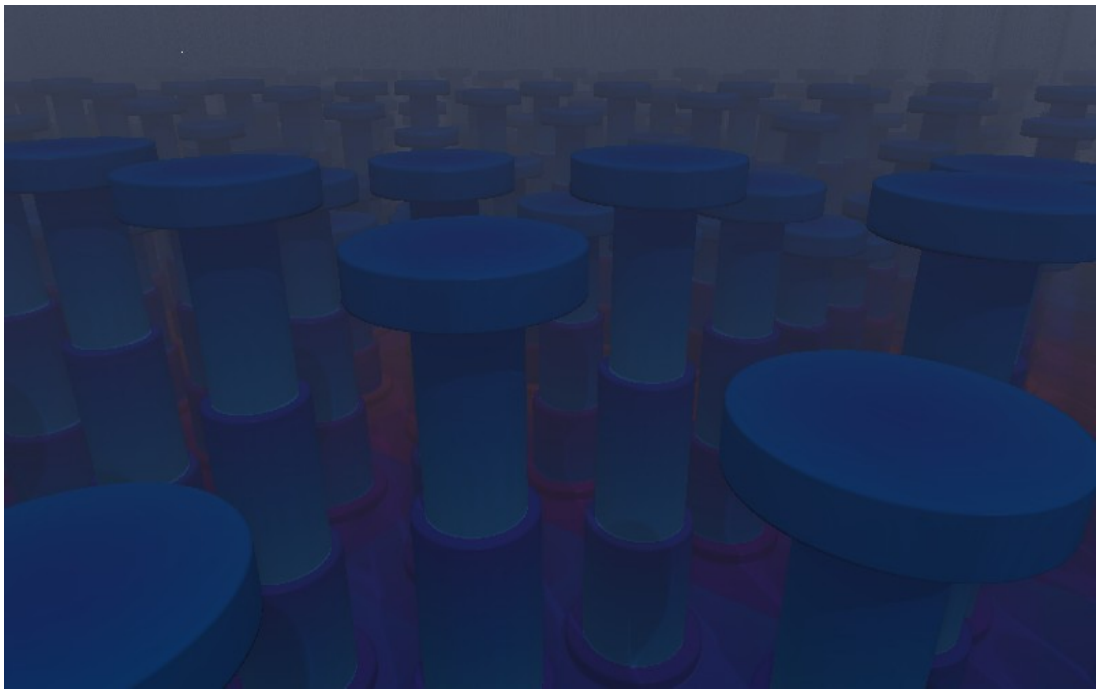


Figure 28: Piston field, default parameters (3 iterations).

Parameters (app defaults). Iterations 3 (range 1–4); step factor 0.7; bounding size 2; *unbounded* (no bounding sphere assumed).

Parameter	Key	Default	Range
Spacing	`cell`	0.7	0.3 … 2
Radius	`radius`	0.22	0.05 … 0.48
Rest height	`base`	0.8	0 … 3
Stroke	`stroke`	0.6	0 … 2
Phase	`phase`	0	0 … 1
Wave	`wave`	1	0 … 3
Cap	`cap`	0.5	0 … 1

7.5 Clockwork

Gears carrying rings of smaller, *meshing* gears. Generation i has $n_i = \max(\mathrm{round}(T/s^i), 6)$ teeth, direction $\delta_i = (-1)^i$ and speed $v_i = s^i$. With $a = \mathrm{atan2}(z_z, z_x) + \mathrm{spin}\,\delta_i v_i$, $r = \|\mathbf{z}_{xz}\|$, $R = 1$ (the frame is rescaled each step), tooth depth t, thickness ϑ:

$$\mathrm{rim} = r - \Big(R - t + t\big(\tfrac{1}{2} + \tfrac{1}{2}\,\mathrm{clamp}(2.5\cos(n_i a), -1, 1)\big)\Big), \tag{66}$$

$$\mathrm{hole} = \max\Big(0.3R - r,\ r - (0.3 + 0.42\,\mathrm{win})R,\ 0.09R - \Delta a\, r\Big), \tag{67}$$

$$\Delta a = \Big|\big((a + \tfrac{\varsigma_5}{2}) \bmod \varsigma_5\big) - \tfrac{\varsigma_5}{2}\Big|, \quad \varsigma_5 = \tfrac{2\pi}{5} \quad \text{(five spokes)}, \tag{68}$$

$$g = \max\Big(0.6\max\big(\max(\mathrm{rim}, -\mathrm{hole}),\ 0.05R - r\big),\ |z_y| - \vartheta\Big), \tag{69}$$

$$g \leftarrow \min\big(g,\ \max(r - 0.09R,\ |z_y| - 1.8\vartheta)\big) \quad \text{(axle)}, \tag{70}$$

and $d \leftarrow \min(d, g/\mathrm{dr})$. Children: fold the azimuth into one of m sectors ($\varsigma = 2\pi/m$), then

$$\mathbf{z} \leftarrow s\,\mathsf{R}_A\big(\mathbf{z} - (R + R/s - t,\ \mathrm{lift},\ 0)\big), \qquad \mathrm{dr} \leftarrow s\,\mathrm{dr}. \tag{71}$$

Spinning the big gear turns every generation at its gear ratio.

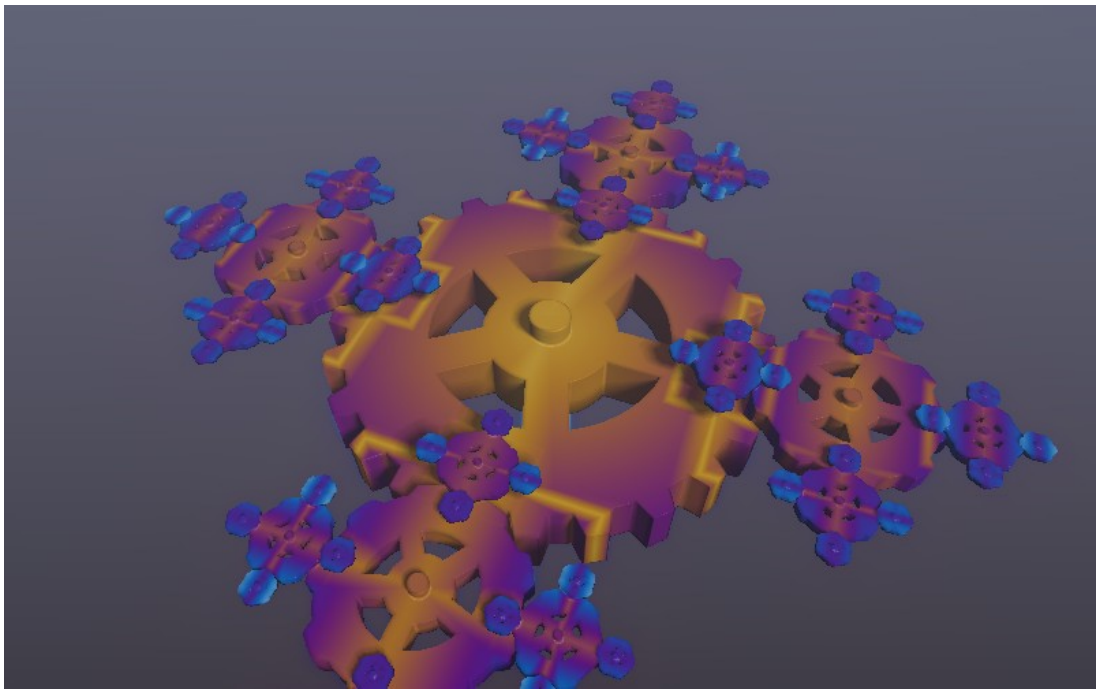


Figure 29: Clockwork, default parameters (4 iterations).

Parameters (app defaults). Iterations 4 (range 1–6); step factor 0.6; bounding size 1.8.

Parameter	Key	Default	Range
Teeth	`teeth`	24	6 … 48
Ratio	`ratio`	2.6	1.8 … 4
Gears per ring	`around`	4	2 … 8
Tooth depth	`tooth`	0.09	0.02 … 0.25
Thickness	`thick`	0.12	0.03 … 0.5
Stack	`lift`	0.14	−0.5 … 0.5
Windows	`windows`	0.6	0 … 1
Spin	`spin`	0°	−180 … 180
Tilt X	`rotX`	0°	−180 … 180
Tilt Y	`rotY`	0°	−180 … 180
Tilt Z	`rotZ`	0°	−180 … 180

7.6 Truss

Girder frames with diagonal braces, each holding smaller frames at its corners. With half-extents $\mathbf{b} = (L, \frac{1}{2}, \frac{1}{2})$ (L = stretch) and beam half-thickness e the frame distance (box-frame SDF) is

$$\mathbf{p}' = |\mathbf{z}| - \mathbf{b}, \quad \mathbf{q}' = |\mathbf{p}' + e| - e, \tag{72}$$

$$d_{\text{frame}} = \min\Big(g(p'_x, q'_y, q'_z),\ g(q'_x, p'_y, q'_z),\ g(q'_x, q'_y, p'_z) \Big), \quad g(\mathbf{u}) = \|\max(\mathbf{u}, 0)\| + \min(\max(u_x, u_y, u_z), 0), \tag{73}$$

plus three diagonal rods (capsule distance, radius $0.7\,e\,\beta$) in the $|\mathbf{z}|$ octant when the brace amount $\beta > 0$. Per iteration:

$$d \leftarrow \min\big(d,\ d_{\text{frame}}/\mathrm{dr}\big); \quad \mathbf{z} \leftarrow |\mathbf{z}|\,,\ \text{sort } y \geq z; \quad \mathbf{z} \leftarrow s\,\mathsf{R}_A\mathbf{z} - \mathbf{o}\,(s-1) \odot (2L, 1, 1); \quad \mathrm{dr} \leftarrow s\,\mathrm{dr}. \tag{74}$$

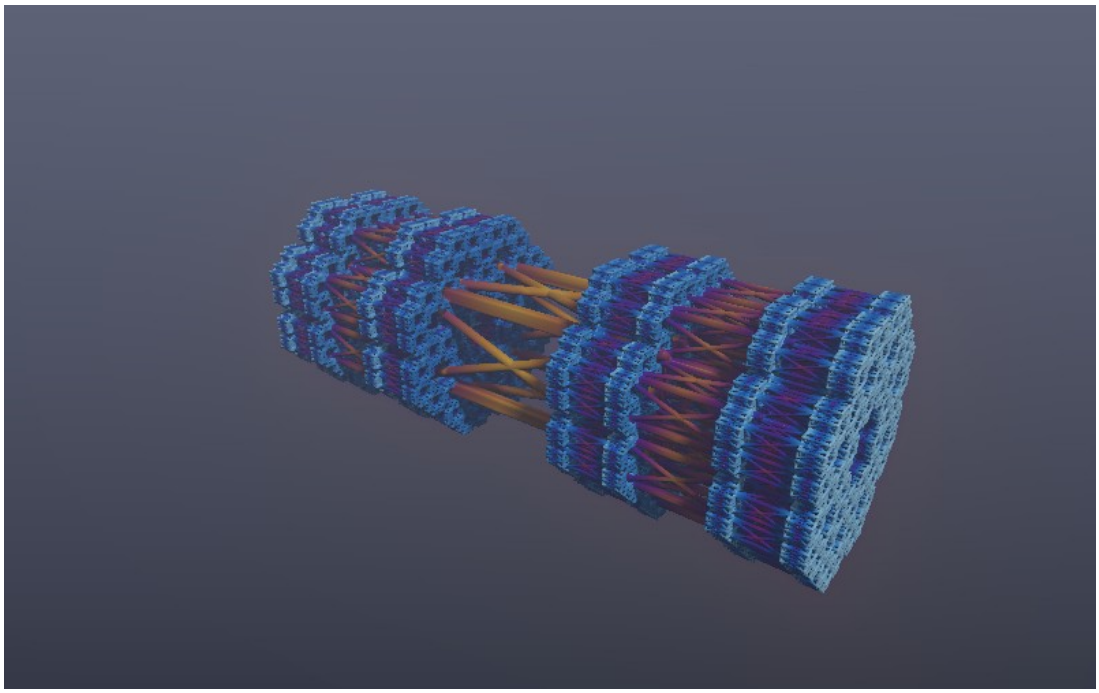


Figure 30: Truss, default parameters (4 iterations).

Parameters (app defaults). Iterations 4 (range 1–7); step factor 0.8; bounding size 1.6.

Parameter	Key	Default	Range
Scale	`scale`	2.6	1.8 . . . 4
Offset X	`offX`	1	0 . . . 2
Offset Y	`offY`	1	0 . . . 2
Offset Z	`offZ`	0.4	0 . . . 2
Beam	`beam`	0.05	0.01 . . . 0.2
Braces	`brace`	1	0 . . . 1
Stretch	`stretch`	1.4	0.4 . . . 3
Rotation X	`rotX`	0°	−180 . . . 180
Rotation Y	`rotY`	0°	−180 . . . 180
Rotation Z	`rotZ`	0°	−180 . . . 180

7.7 Reactor

Finned rings on spokes, each carrying smaller rings tilted on their own axes. Ring of radius 1 with rectangular section (half-width w, half-depth dp); $a = \operatorname{atan2}(z_z, z_x) + \text{spin}$, $r = \|\mathbf{z}_{xz}\|$:

$$\mathbf{q} = |(r-1,\ z_y)| - (w, \text{dp}), \qquad d_{\text{ring}} = \|\max(\mathbf{q}, 0)\| + \min(\max(q_x, q_y), 0), \tag{75}$$

$$d_{\text{fin}} = \max\Big(\text{rect}\big(\,|(r-1, z_y)| - (1.8w,\ 0.85\,\text{dp})\big),\ \Delta_F\, r - 0.012\Big), \tag{76}$$

$$F \text{ fins; } \Delta_F = \text{angular distance to the nearest fin}, \tag{77}$$

$$d_{\text{ring}} \leftarrow \min(d_{\text{ring}},\ 0.8\, d_{\text{fin}}), \tag{78}$$

and, for $S > 0$ spokes, a spoke capsule and a hub $\|(r, z_y)\| - 1.4w$ are added with min. Children: fold the azimuth into one of m sectors, move to the ring's rim ($z_x \leftarrow z_x - 1$), tilt by φ in the (y, z) plane and scale:

$$\mathbf{z}_{yz} \leftarrow \begin{pmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix} \mathbf{z}_{yz}, \quad \mathbf{z} \leftarrow s\mathbf{z}, \quad \text{dr} \leftarrow s\,\text{dr}. \tag{79}$$

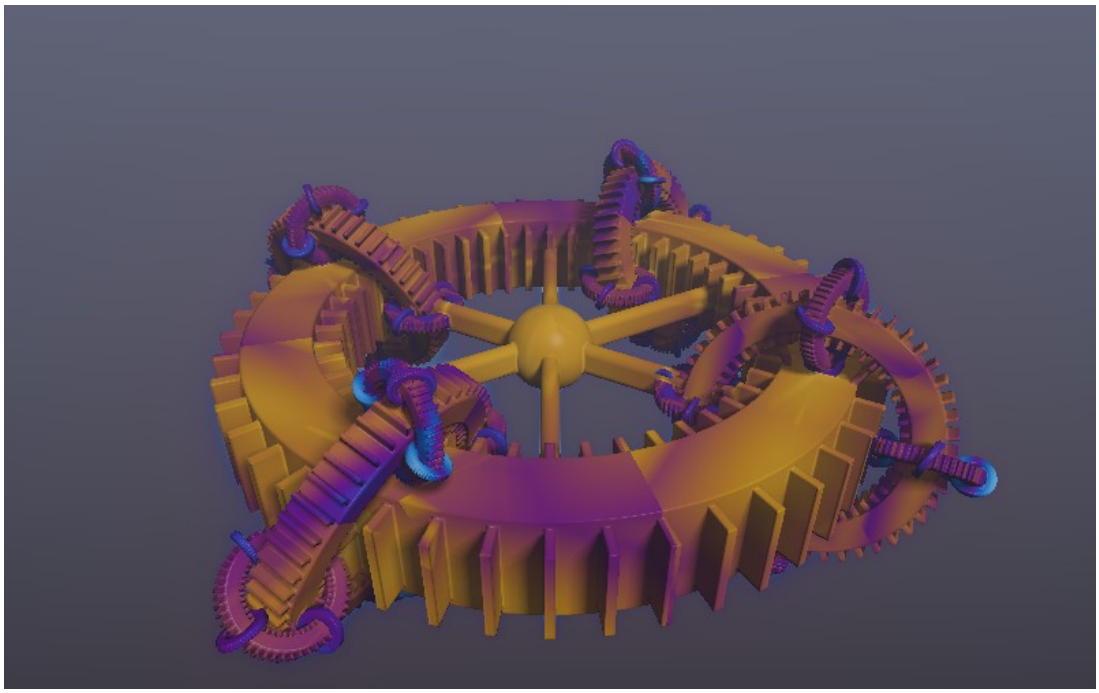


Figure 31: Reactor, default parameters (4 iterations).

Parameters (app defaults). Iterations 4 (range 1–6); step factor 0.7; bounding size 1.6.

Parameter	Key	Default	Range
Ratio	`ratio`	2.7	1.8 . . . 4
Rings per ring	`around`	4	2 . . . 8
Spokes	`spokes`	6	0 . . . 12
Fins	`fins`	48	0 . . . 96
Ring width	`width`	0.12	0.03 . . . 0.4
Ring depth	`depth`	0.18	0.02 . . . 0.5
Tilt	`tilt`	70°	0 . . . 90
Spin	`spin`	0°	−180 . . . 180

8 Landscapes (heightfields)

These are not volumetric fractals: each defines a height $h(x, z)$ and returns a scaled vertical distance $\mathrm{DE} = \lambda\,(y - h)$, with a safety factor $\lambda < 1$ because the slope can exceed one.

Value noise. hash($\mathbf{v}$) as in Skyline (constants $234.34, 435.345, 34.23$ here); with $\mathbf{i} = \lfloor \mathbf{q} \rfloor$, $\mathbf{f} = \mathrm{fract}(\mathbf{q})$, $\mathbf{u} = \mathbf{f}^2(3 - 2\mathbf{f})$:

$$n(\mathbf{q}) = \mathrm{lerp}\big(\mathrm{lerp}(h_{00}, h_{10}, u_x),\ \mathrm{lerp}(h_{01}, h_{11}, u_x),\ u_y\big). \tag{80}$$

The octave map is $\mathbf{q} \leftarrow M\mathbf{q}$ with $M = \begin{pmatrix} 1.6 & -1.2 \\ 1.2 & 1.6 \end{pmatrix}$ (a rotation by 36.9° and scaling by 2).

8.1 Folded terrain

Ridged fractional Brownian motion with domain warping. $\mathbf{q}_0 = f\,\mathbf{p}_{xz} + 13.1 \cdot \mathrm{seed}$, ridge amount $\rho \in [0, 1]$, roughness (gain) G, warp W:

$$n_k = (1 - \rho)\,n(\mathbf{q}_k) + \rho\,\big(1 - |2n(\mathbf{q}_k) - 1|\big), \qquad \Sigma_k = \Sigma_{k-1} + a_k n_k, \quad a_k = G^k, \tag{81}$$

$$\mathbf{q}_{k+1} = M\mathbf{q}_k + W\,(\Sigma_k,\ -\Sigma_k), \qquad h_n = \Sigma \Big/ \sum a_k, \tag{82}$$

$$h = \max\big(H\,(h_n - 0.45),\ \mathrm{sea}\big), \qquad \mathrm{DE} = 0.55\,(p_y - h). \tag{83}$$

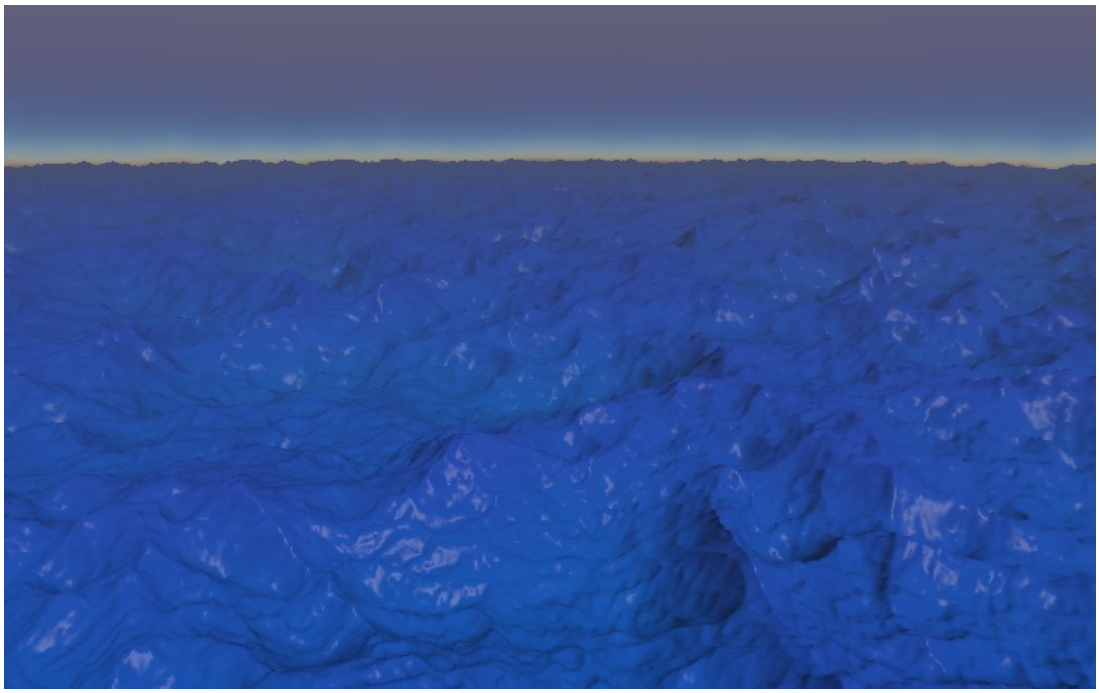


Figure 32: Folded terrain, default parameters (7 iterations).

Parameters (app defaults). Iterations 7 (range 2–10); step factor 0.45; bounding size 3; *unbounded* (no bounding sphere assumed).

Parameter	Key	Default	Range
Feature size	`freq`	0.6	0.1 … 3
Height	`height`	1.4	0 … 4
Ridge	`ridge`	0.7	0 … 1
Roughness	`gain`	0.48	0.25 … 0.7
Warp	`warp`	0.35	0 … 1.5
Sea level	`sea`	−0.35	−1 … 1
Seed	`seed`	3	0 … 100

8.2 Kali coral plain

Ground grown from Kali's abs-inversion set, tiled by mirroring so it repeats without seams:

$$\mathbf{q}_0 = 1.2\left|\,2\,\mathrm{fract}\big(\tfrac{f}{4}\,\mathbf{p}_{xz}+\tfrac{1}{2}\big)-1\,\right|, \qquad \mathbf{q}_{k+1}=\frac{|\mathbf{q}_k|}{\max(\|\mathbf{q}_k\|^2,10^{-3})}-\mathbf{c},\quad \mathbf{c}=(c_x,c_y), \tag{84}$$

$$\mathrm{acc}=\sum_k \exp\big(-\gamma\,\big|\,\|\mathbf{q}_k\|-\|\mathbf{q}_{k-1}\|\,\big|\big), \qquad h_n=\mathrm{clamp}(\mathrm{acc}/N,0,1), \tag{85}$$

$$h=\max(H\,h_n,\ \mathrm{floor}), \qquad \mathrm{DE}=0.4\,(p_y-h). \tag{86}$$

γ = *Contrast*.

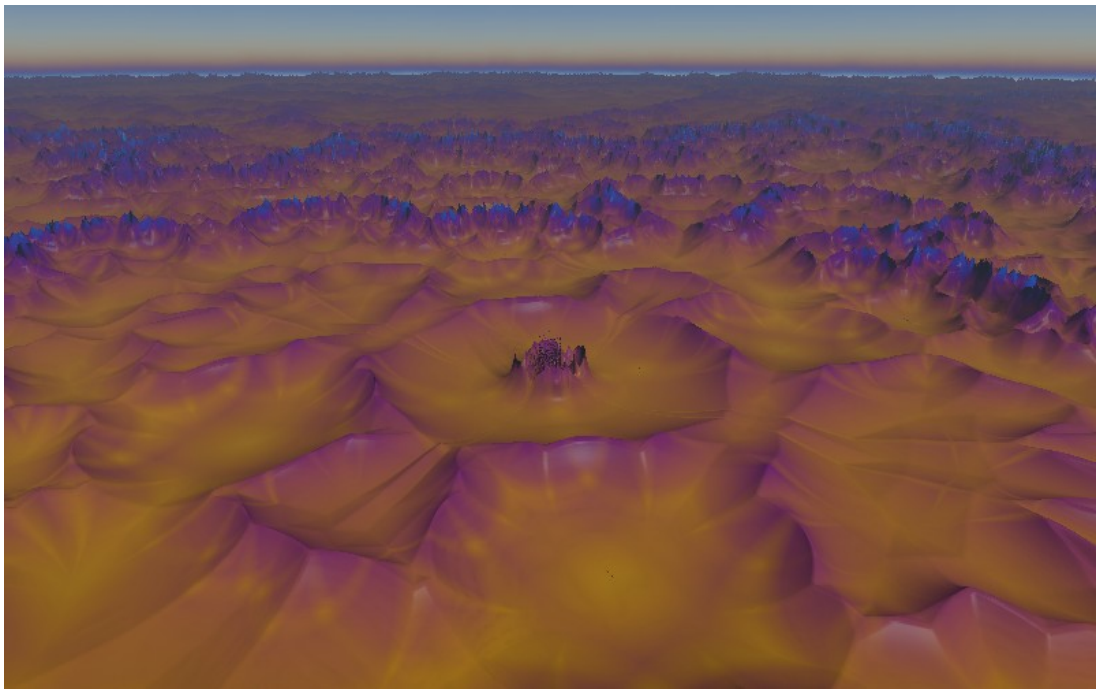


Figure 33: Kali coral plain, default parameters (9 iterations).

Parameters (app defaults). Iterations 9 (range 3–20); step factor 0.35; bounding size 3; *unbounded* (no bounding sphere assumed).

Parameter	Key	Default	Range
Feature size	`freq`	0.12	0.02 . . . 2
Height	`height`	0.7	0 . . . 3
C x	`cx`	0.58	0 . . . 1.5
C y	`cy`	0.73	0 . . . 1.5
Contrast	`focus`	1.5	0.5 . . . 8
Floor	`floor`	−0.2	−1 . . . 1

8.3 Eroded ridges

Noise whose small detail fades on steep slopes, as erosion leaves it. Uses value noise *with analytic derivatives* $(n,\ \partial_x n,\ \partial_z n)$. With $\mathbf{D}_k = \sum_{j \le k} \nabla n(\mathbf{q}_j)$ the accumulated gradient, erosion η, gain G:

$$a = \sum_k b_k \, \frac{n(\mathbf{q}_k)}{1 + \eta \, \|\mathbf{D}_k\|^2}, \qquad b_k = G^k, \qquad h_n = a \Big/ \sum_k b_k, \tag{87}$$

$$h_n' = h_n - 0.6\,\kappa\,\mathrm{smoothstep}(0.35,\ 0.15,\ h_n) \quad \text{(canyon amount } \kappa\text{)}, \tag{88}$$

$$h = H\,(h_n' - 0.3), \qquad \mathrm{DE} = 0.5\,(p_y - h). \tag{89}$$

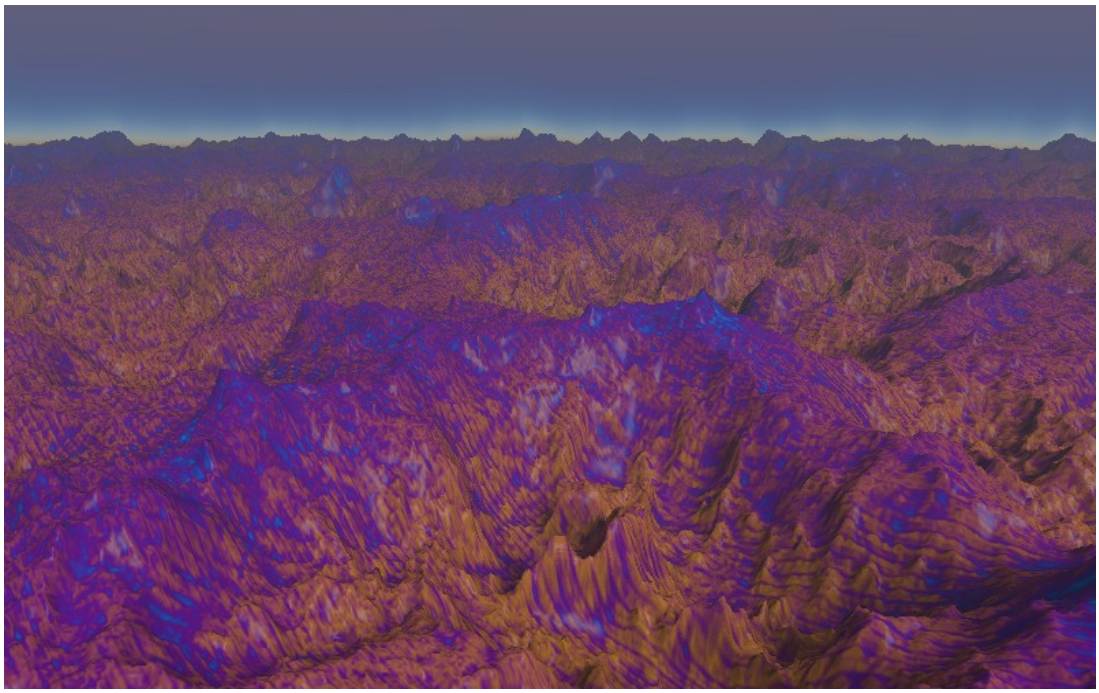


Figure 34: Eroded ridges, default parameters (8 iterations).

Parameters (app defaults). Iterations 8 (range 2–10); step factor 0.45; bounding size 3; *unbounded* (no bounding sphere assumed).

Parameter	Key	Default	Range
Feature size	`freq`	0.45	0.1 . . . 3
Height	`height`	1.8	0 . . . 4
Erosion	`erosion`	1.2	0 . . . 3
Roughness	`gain`	0.5	0.25 . . . 0.7
Canyon	`canyon`	0.3	0 . . . 1
Seed	`seed`	11	0 . . . 100

A Quick reference: which equation is which

Family	Formulas	Distance estimate
Escape-time	Mandelbulb, Quaternion Julia, Folded bulb hybrid	$\frac{1}{2} r \ln r / \mathrm{dr}$ (potential)
Box/sphere fold	Mandelbox, Amazing Box, Vaulted box, Amazing Surf	$\lVert \mathbf{z} \rVert \,/\, \lvert \mathrm{dr} \rvert$
Abs-inversion	KaliBox, Pseudo-Kleinian, Apollonian	scale-corrected
KIFS	Menger, Sierpinski, Octahedral, Kaleidoscopic, Icosahedral	$(\lVert \mathbf{z} \rVert - 1)/s^N$ or $\mathrm{sdBox}\,/s^N$
Hyperbolic	Hyperbolic honeycomb	d/sc in Poincaré ball
Branching	Tree, Dendrite, Nerve cell, Root, Mycelium, Coral	$\min_k \mathrm{cap}_k / \mathrm{dr}_k$
Procedural/IFS	Ziggurat, Skyline, Turbine, Greeble, Pipeworks, Pistons, Clockwork, Truss, Reactor	$\min_k(\text{primitive})/\mathrm{dr}$
Heightfield	Folded terrain, Kali coral plain, Eroded ridges	$\lambda(y - h)$